Faculté des science techniques



INSTITUT UNIVERSITAIRE DES SCIENCES

LICENCE 3

PREPARER PAR:

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PROF: ISMAEL SAINT-AMOUR

DATE: 07 NOVEMBRE 2026

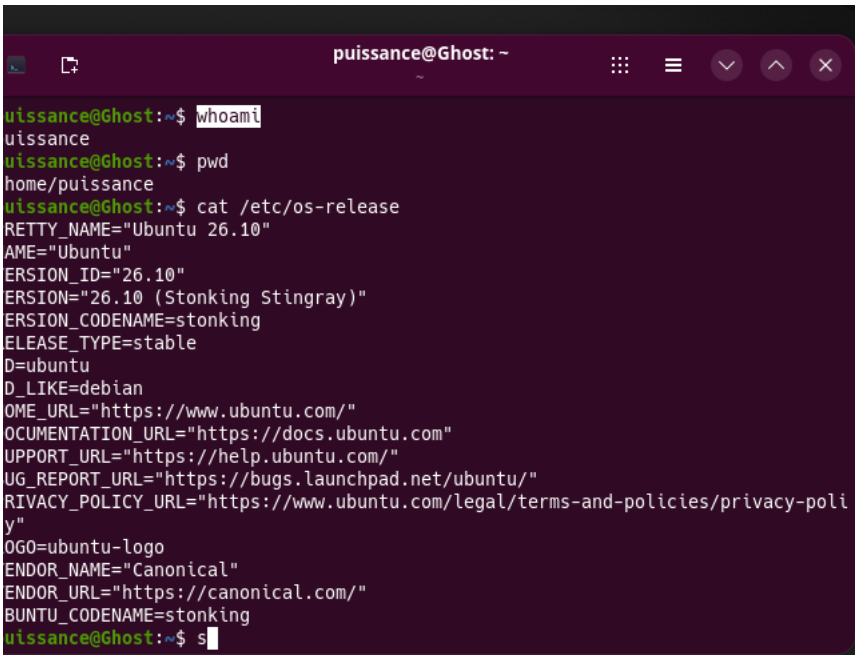
Ojectif du TD:

Ce travail pratique vise à me permettre d'apprendre à gérer les fichiers et les dossiers à l'aide des commandes Linux, à effectuer des recherches et des manipulations de texte, à surveiller les ressources et les processus du système, à gérer les logiciels avec APT (installation et désinstallation), ainsi qu'à découvrir et comparer les environnements de bureau KDE Plasma et XFCE.

Démarche suivie

Réalisation des quatre premiers exercices dans le répertoire ~/td_linux, en exécutant les commandes une par une et en prenant des captures d'écran aux étapes essentielles.
Mise en pratique des commandes de la section « Exécutez ces commandes », concernant le réseau, les journaux système et quelques outils de divertissement.
Création d'un instantané de la machine virtuelle avant l'installation des environnements de bureau, examen des simulations APT, puis installation de KDE Plasma et XFCE.
Vérification du fonctionnement des deux environnements à travers le terminal, le gestionnaire de fichiers, les documents du TD, le fond d'écran, le lanceur d'applications et les paramètres, ainsi que comparaison de leur consommation mémoire avec la commande free -h.

Exercice 1 – Se repérer et manipuler des fichiers
Question 1 : utilisateur, répertoire courant, distribution



Expliquer la différence entre un chemin absolu et un chemin relatif

Le chemin absolu part toujours de la racine (/), tandis que le chemin relatif part du répertoire dans lequel on se trouve actuellement.

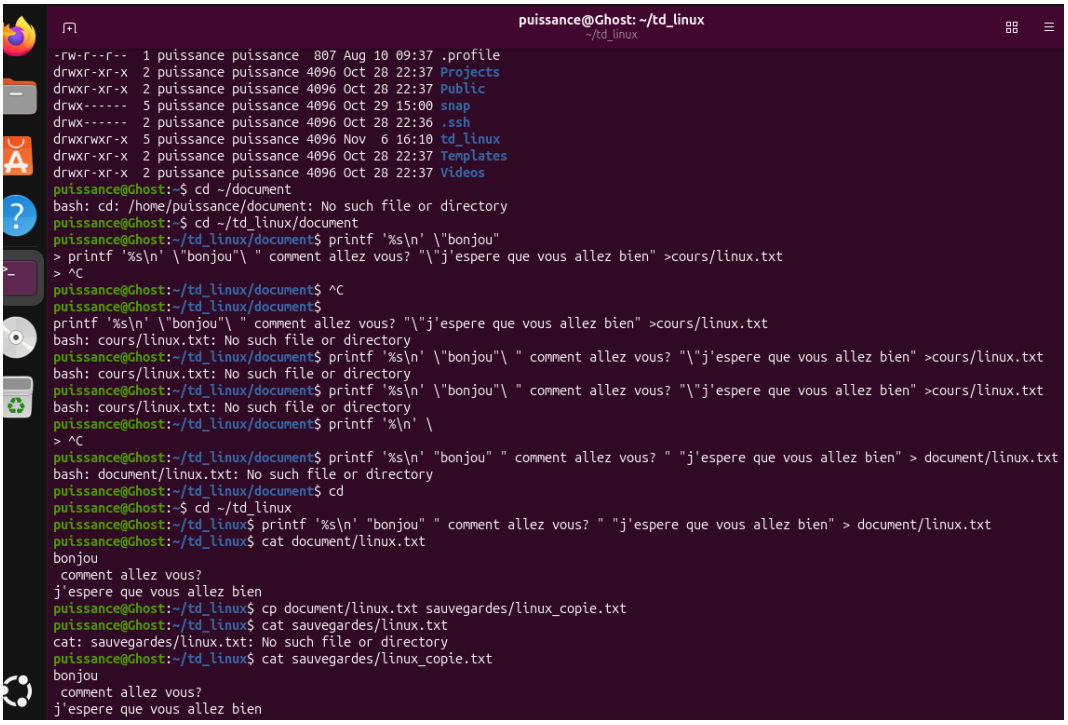
Capture : whoami (david), pwd (/home/david) et cat /etc/os-release (Ubuntu 26.04.1 LTS).

Un chemin absolu commence par / (la racine) et désigne toujours le même

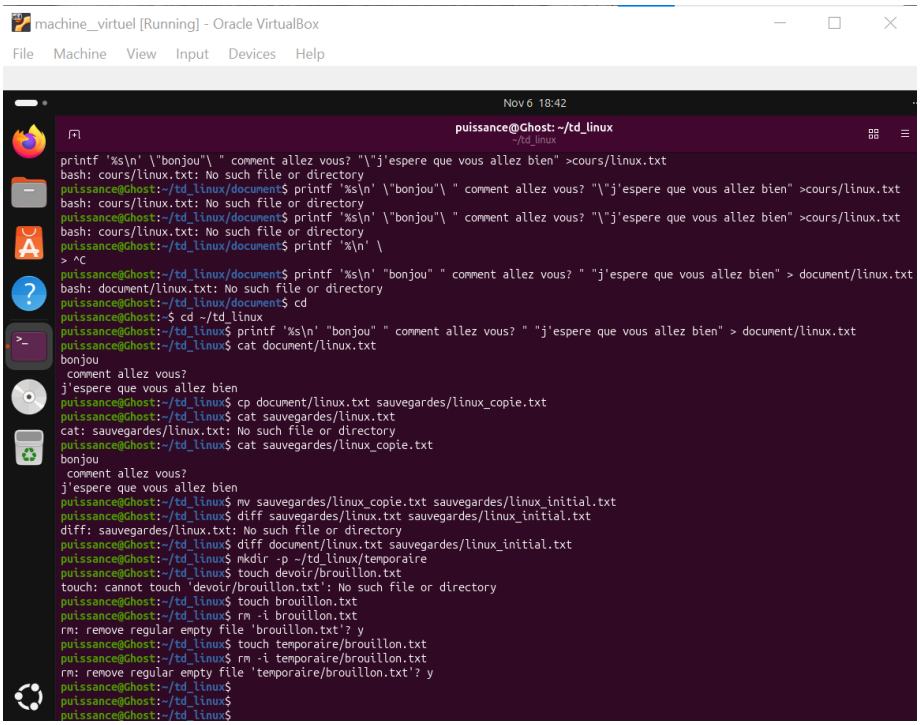
emplacement, quel que soit le répertoire courant tandisque pour un chemin relatif ,son

change selon le dossier dans lequel on se trouve.

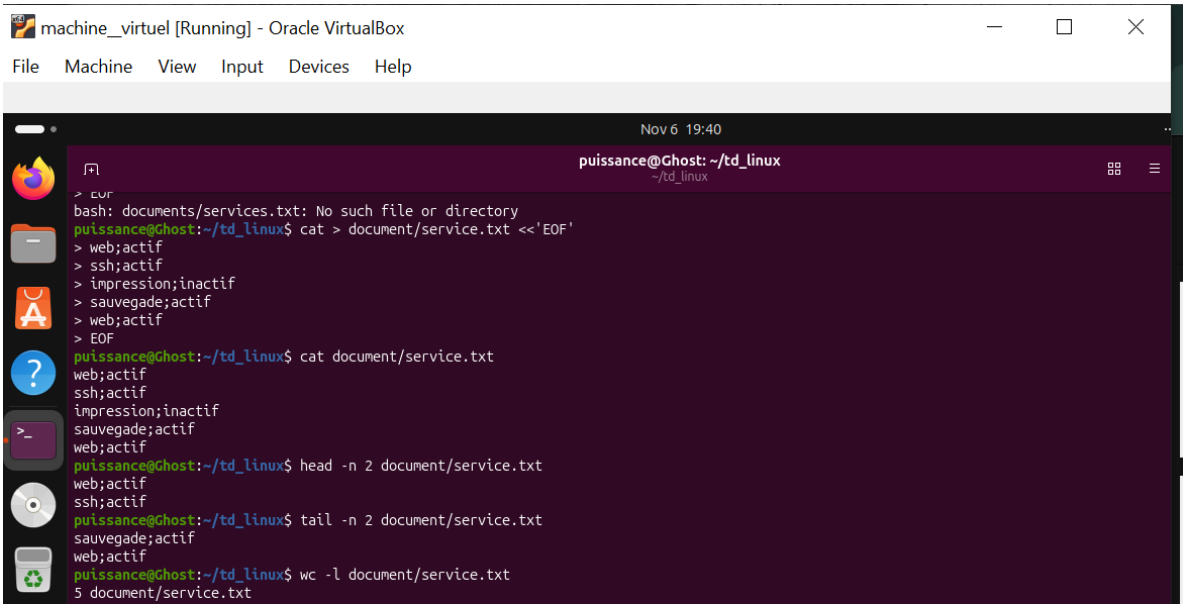
II creation de l'arborecence et notes.txt,copie et renommage



Suite : copie et renommage de notes.txt



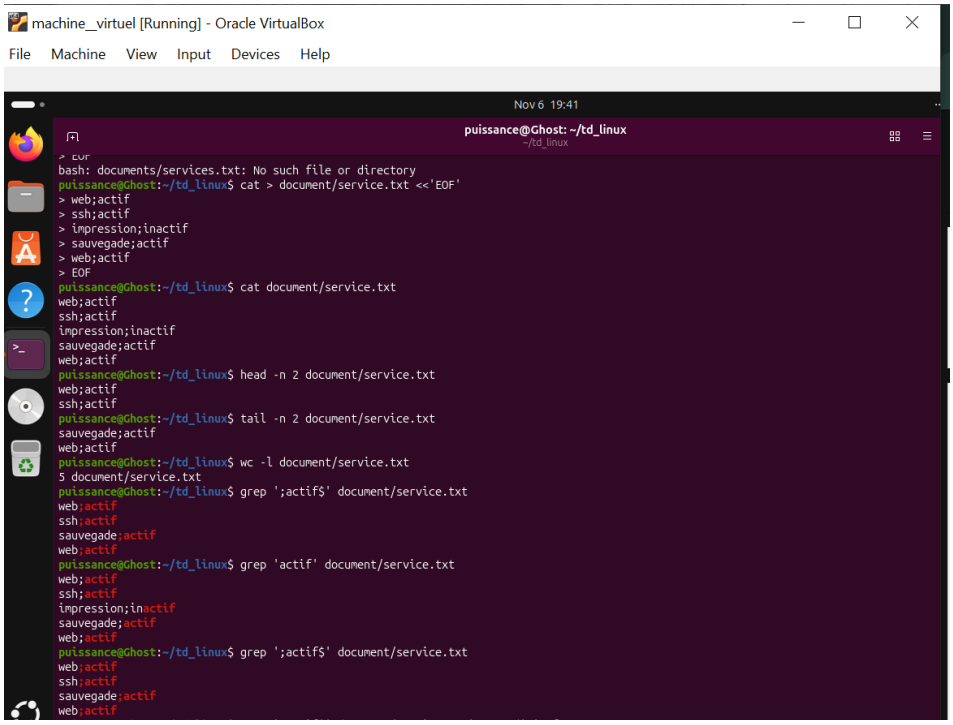
EXERCICE2 – Utilitaires et traitement de texte Creation du fichier services.txt et afficher le fichier



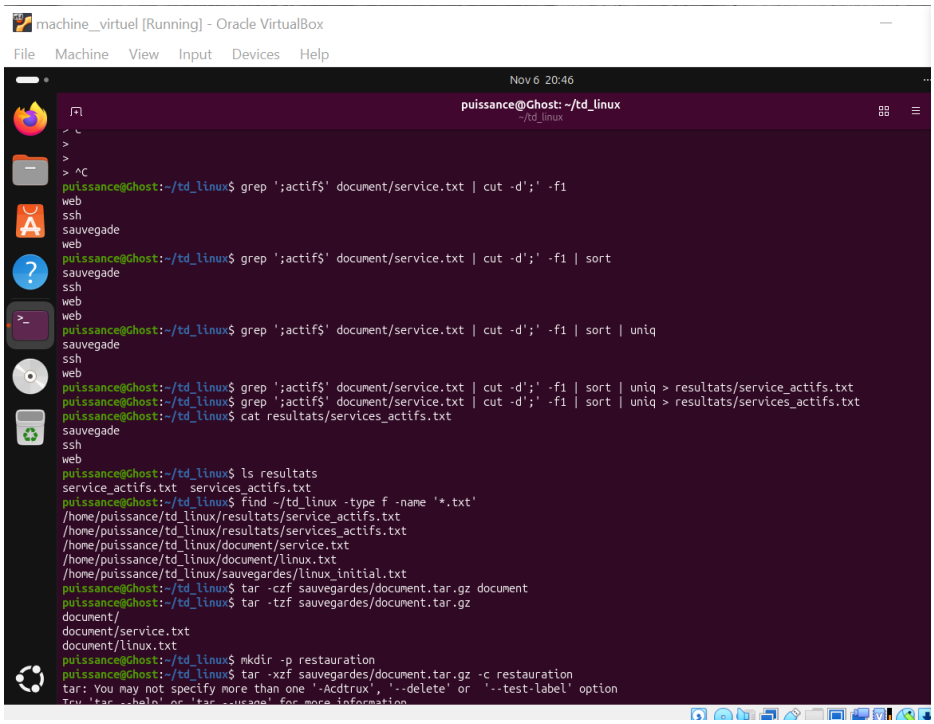
Expliquer la différence entre les opérateurs > et >>

> sert à remplacer le contenu d'un fichier, tandis que >> sert à ajouter du contenu sans supprimer ce qui existe déjà.

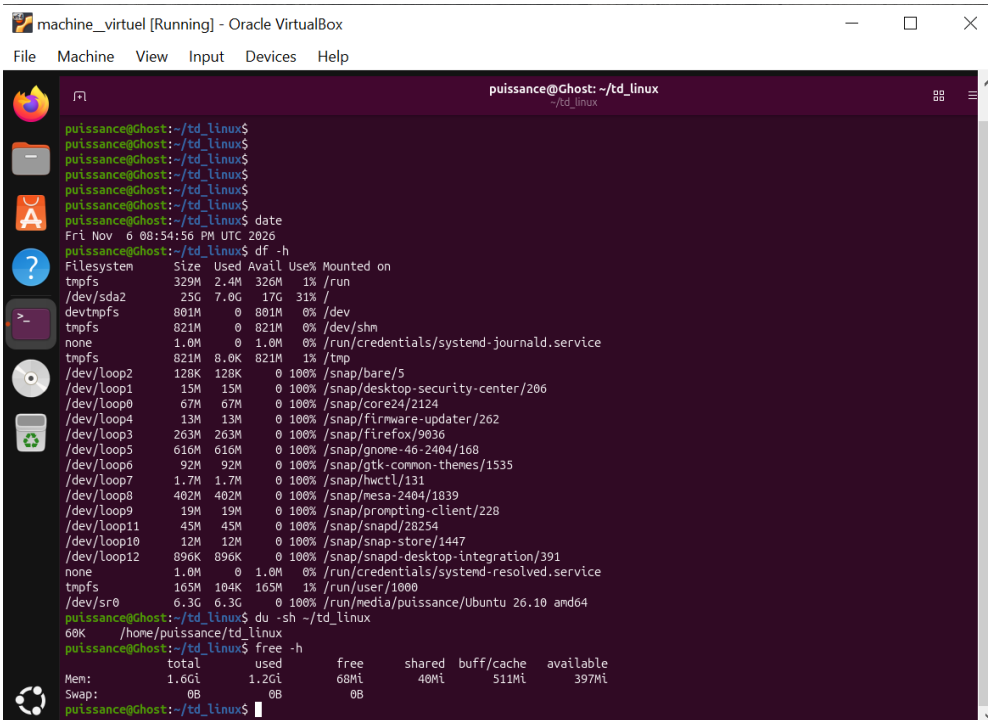
2 – Extraire uniquement lignes dont le deuxieme champ vaut << actif>>.



3 et 4. Exécuter une seule chaîne de commandes et rechercher les fichiers .txt


machine_virtuel [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Nov 6 20:46
puissance@Ghost: ~/td_linux
~/td_linux
> ^C
puissance@Ghost:~/td_linux$ grep ';actif$' document/service.txt | cut -d';' -f1
web
ssh
sauvegade
web
puissance@Ghost:~/td_linux$ grep ';actif$' document/service.txt | cut -d';' -f1 | sort
sauvegade
ssh
web
web
puissance@Ghost:~/td_linux$ grep ';actif$' document/service.txt | cut -d';' -f1 | sort | uniq
sauvegade
ssh
web
puissance@Ghost:~/td_linux$ grep ';actif$' document/service.txt | cut -d';' -f1 | sort | uniq > resultats/service_actifs.txt
puissance@Ghost:~/td_linux$ grep ';actif$' document/service.txt | cut -d';' -f1 | sort | uniq > resultats/services_actifs.txt
puissance@Ghost:~/td_linux$ cat resultats/services_actifs.txt
sauvegade
ssh
web
puissance@Ghost:~/td_linux$ ls resultats
service_actifs.txt services_actifs.txt
puissance@Ghost:~/td_linux$ find ~/td_linux -type f -name '*.txt'
/home/puissance/td_linux/resultats/service_actifs.txt
/home/puissance/td_linux/resultats/services_actifs.txt
/home/puissance/td_linux/document/service.txt
/home/puissance/td_linux/document/linux.txt
/home/puissance/td_linux/sauvegardes/linux_initial.txt
puissance@Ghost:~/td_linux$ tar -czf sauvegardes/document.tar.gz document
puissance@Ghost:~/td_linux$ tar -tzf sauvegardes/document.tar.gz
document/
document/service.txt
document/linux.txt
puissance@Ghost:~/td_linux$ mkdir -p restauration
puissance@Ghost:~/td_linux$ tar -xzf sauvegardes/document.tar.gz -c restauration
tar: You may not specify more than one '-Acdtrux', '--delete' or '--test-label' option

5- achive tar

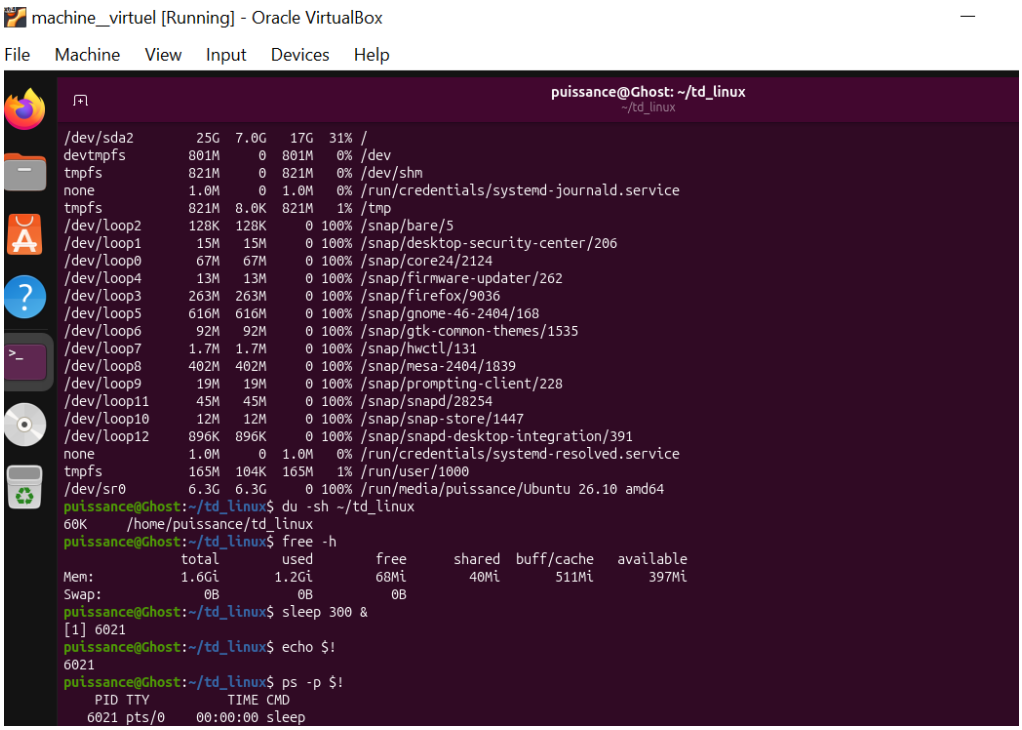

machine_virtuel [Running] - Oracle VirtualBox
File Machine View Input Devices Help
puissance@Ghost: ~/td_linux
~/td_linux
puissance@Ghost:~/td_linux$ date
Fri Nov 6 08:54:56 PM UTC 2026
puissance@Ghost:~/td_linux$ df -h
Filesystem Size Used Avail Use% Mounted on
tmpfs 329M 2.4M 326M 1% /run
/dev/sda2 25G 7.0G 17G 31% /
devtmpfs 801M 0 801M 0% /dev
tmpfs 821M 0 821M 0% /dev/shm
none 1.0M 0 1.0M 0% /run/credentials/systemd-journald.service
tmpfs 821M 8.0K 821M 1% /tmp
/dev/loop2 128K 128K 0 100% /snap/bare/5
/dev/loop1 15M 15M 0 100% /snap/desktop-security-center/206
/dev/loop0 67M 67M 0 100% /snap/core24/2124
/dev/loop4 13M 13M 0 100% /snap/firmware-updater/262
/dev/loop3 263M 263M 0 100% /snap/firefox/9036
/dev/loop5 616M 616M 0 100% /snap/gnome-46-2404/168
/dev/loop6 92M 92M 0 100% /snap/gtk-common-themes/1535
/dev/loop7 1.7M 1.7M 0 100% /snap/hwctl/131
/dev/loop8 402M 402M 0 100% /snap/mesa-2404/1839
/dev/loop9 19M 19M 0 100% /snap/prompting-client/228
/dev/loop11 45M 45M 0 100% /snap/snapd/28254
/dev/loop10 12M 12M 0 100% /snap/snap-store/1447
/dev/loop12 896K 896K 0 100% /snap/snapd-desktop-integration/391
none 1.0M 0 1.0M 0% /run/credentials/systemd-resolved.service
tmpfs 165M 104K 165M 1% /run/user/1000
/dev/sr0 6.3G 6.3G 0 100% /run/media/puissance/Ubuntu 26.10 amd64
puissance@Ghost:~/td_linux$ du -sh ~/td_linux
60K /home/puissance/td_linux
puissance@Ghost:~/td_linux$ free -h
total used free shared buff/cache available
Mem: 1.6Gi 1.2Gi 68Mi 40Mi 511Mi 397Mi
Swap: 0B 0B 0B

EXERCICE 3 – Observation du système et des processus

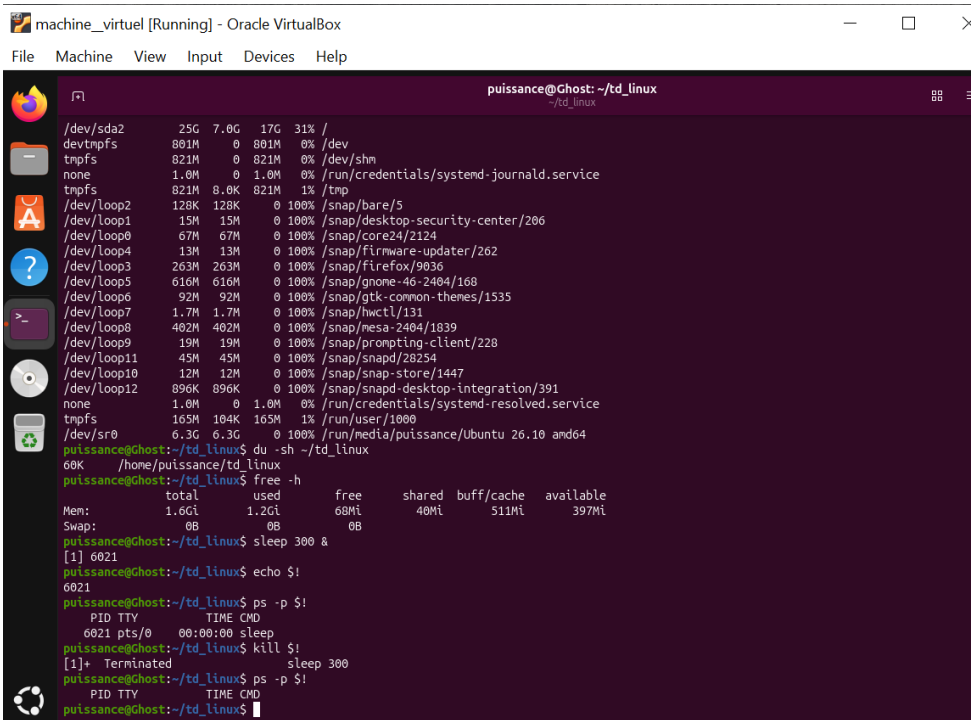
Expliquer la différence entre df et du.

Réponse : df sert à vérifier l'espace disponible sur le disque, tandis que du sert à connaître l'espace occupé par un fichier ou un dossier.

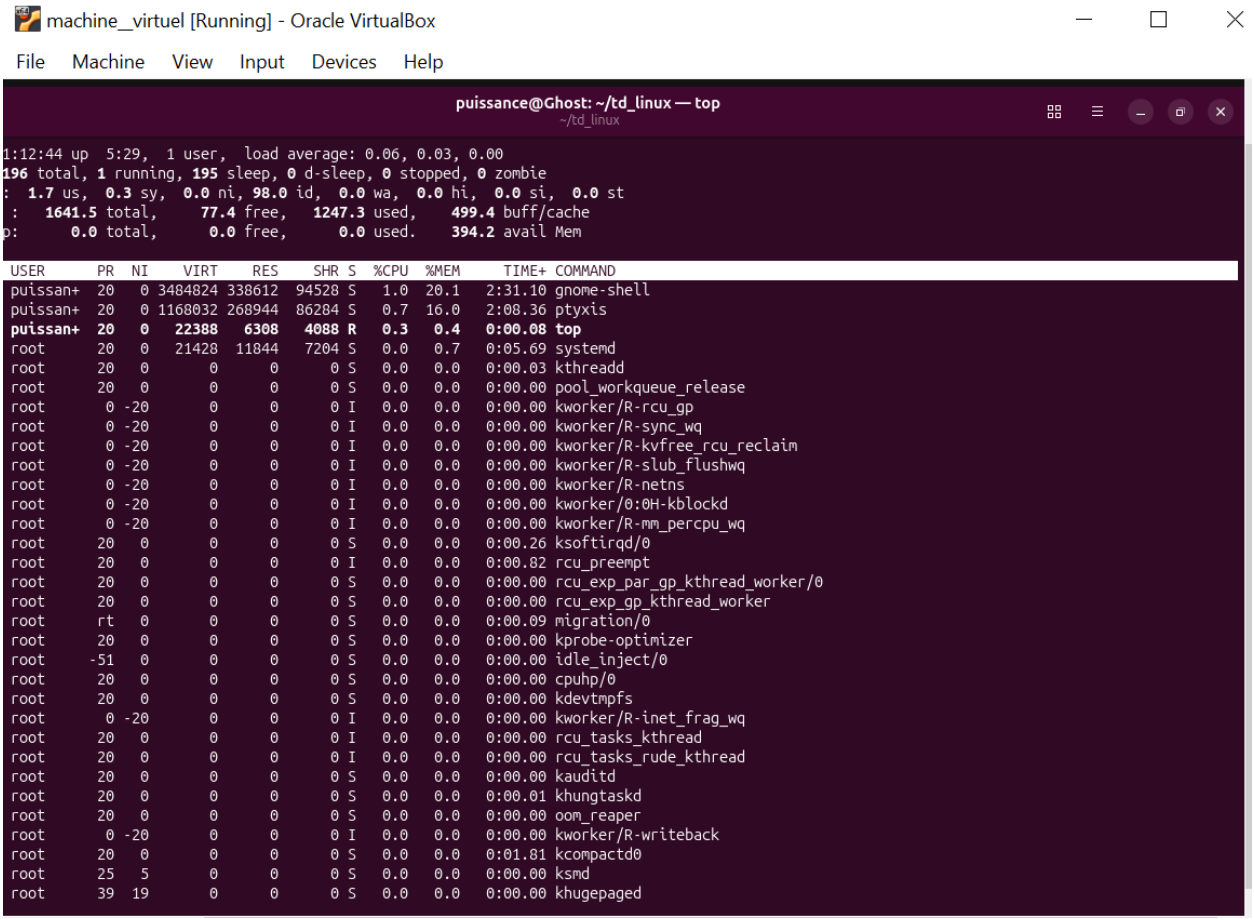
1. Observer le système et les processus



2 et 3. Exécuter un processus en arrière-plan et terminer le processus



4. Observer les processus avec top



EXERCICE 4 – Gestion des paquets

GESTION DES PAQUETS La gestion des paquets est une partie essentielle de l'administration système sur les distributions Linux.

Les paquets (packages) sont des archives de fichiers compressées qui contiennent des logiciels, des bibliothèques, des fichiers de configuration et d'autres éléments nécessaires à l'installation et au fonctionnement d'une application sur un système d'exploitation. Les paquets facilitent la distribution, l'installation, la mise à jour et la suppression de logiciels sur un système informatique.

Dépôt (Repository) : C'est un espace de stockage en ligne qui contient des logiciels et des paquets que l'on peut télécharger et installer sur un ordinateur.

Dépendance (Dependency) : C'est un logiciel ou un paquet nécessaire au fonctionnement ou à l'installation d'un autre logiciel. Exercices 4

Expliquer la différence entre :

```
sudo apt update
```

```
sudo apt upgrade
```

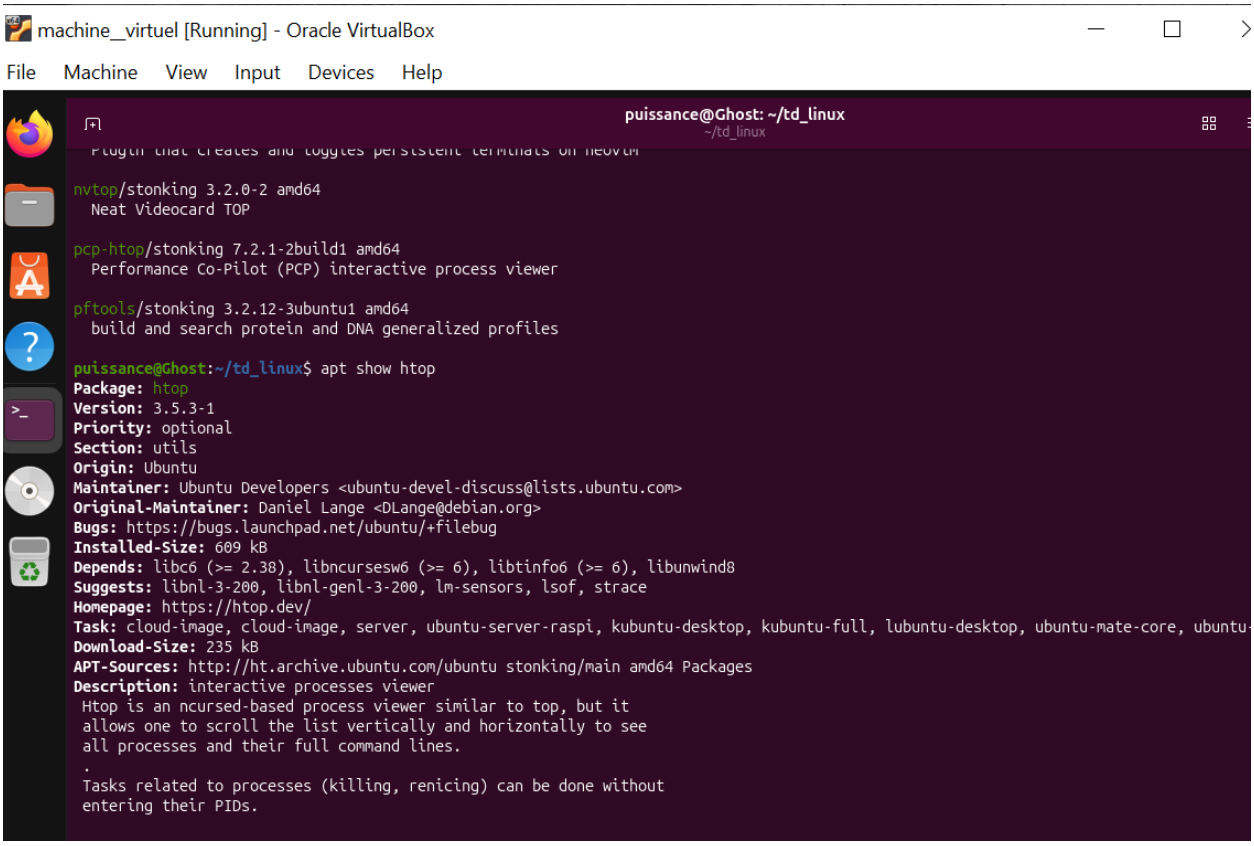
La commande update recherche les mises à jour disponibles, tandis que la commande upgrade installe ces mises à jour.

Quelle commande actualise l'index local ? Laquelle met à niveau les paquets installés ?

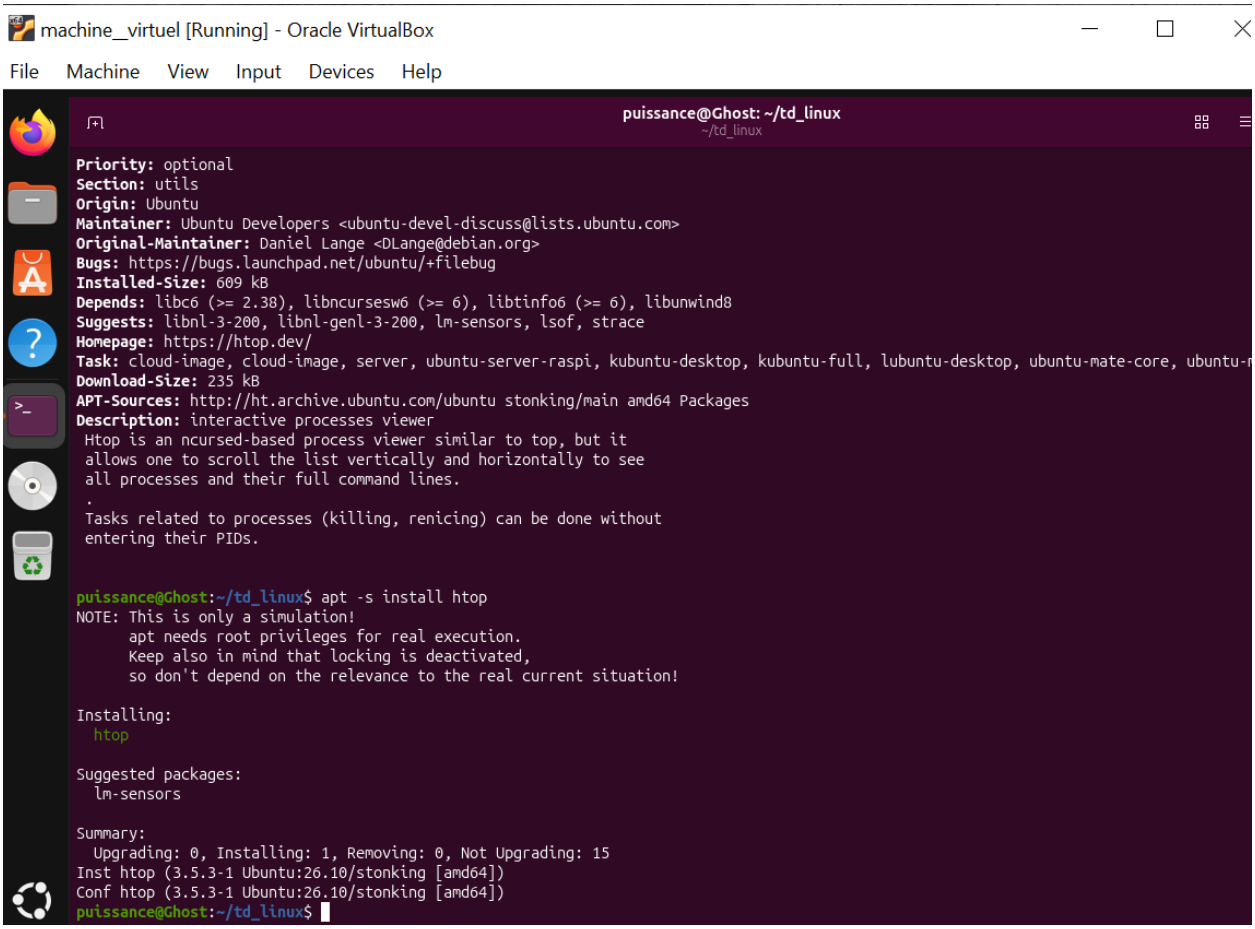
```
apt update : actualise la liste des paquets disponibles.
```

```
apt upgrade : installe les mises à jour des paquets déjà installés.
```

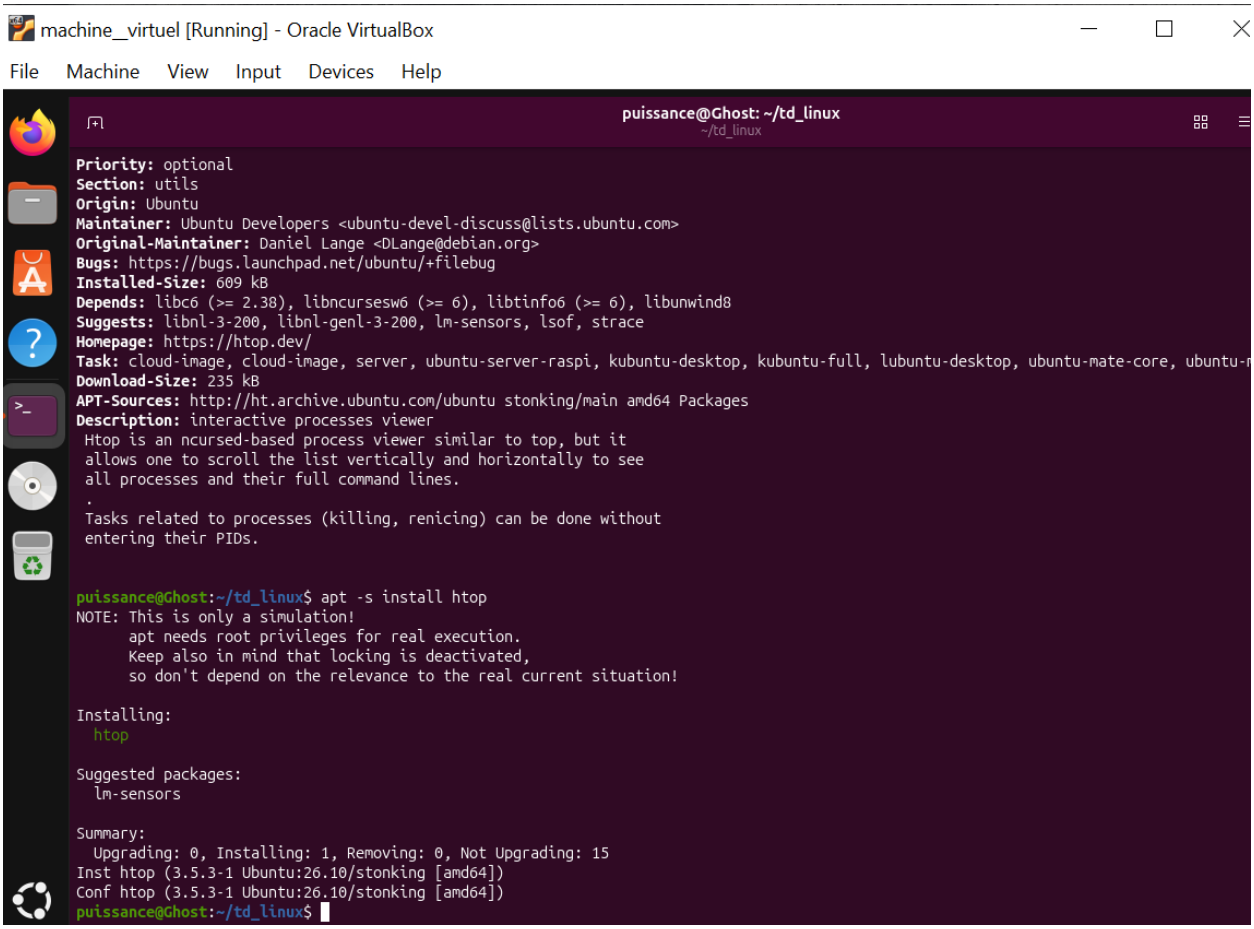
3. Actualiser, rechercher et consulter



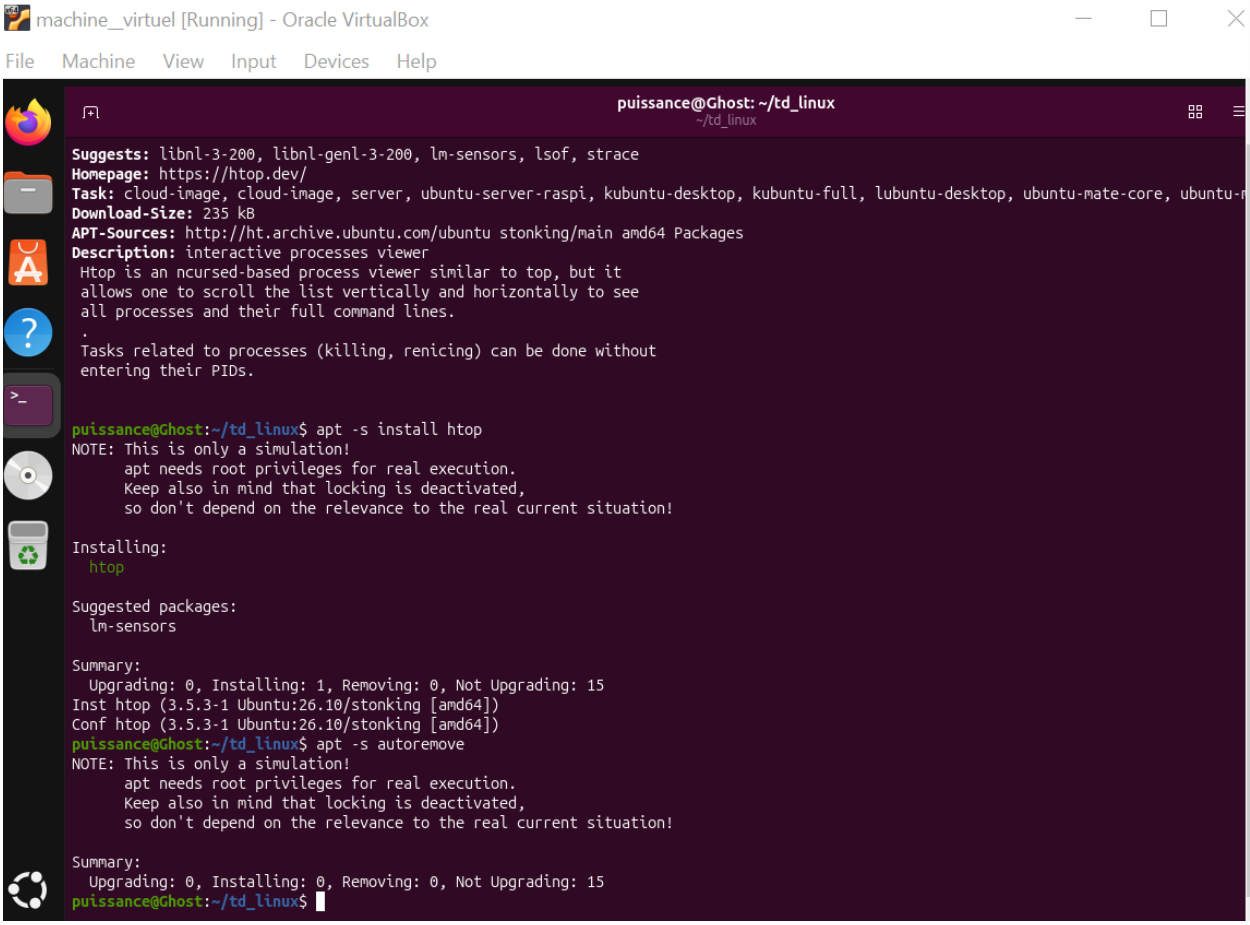
4. Simuler l’installation



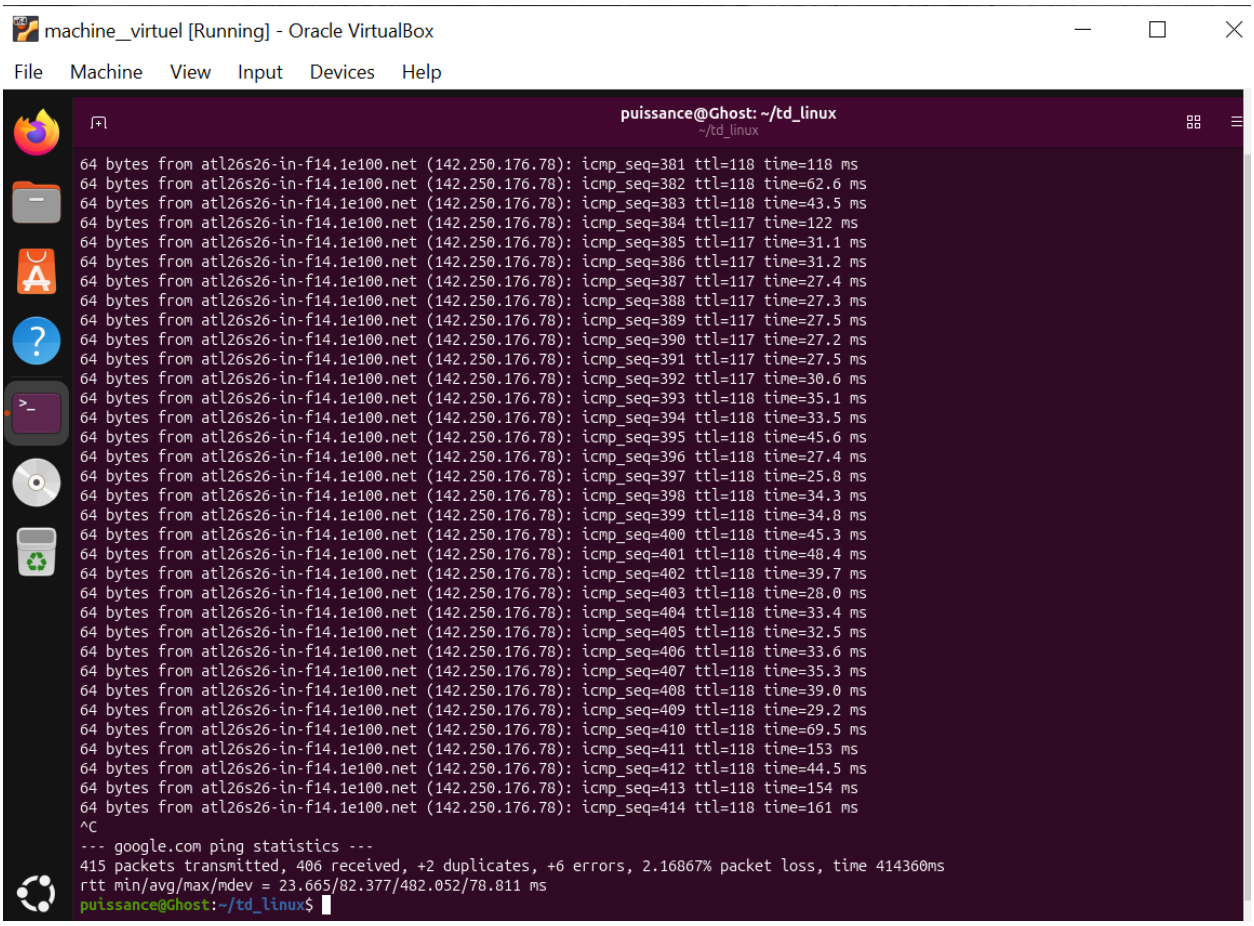
5. Installer, vérifier et lancer

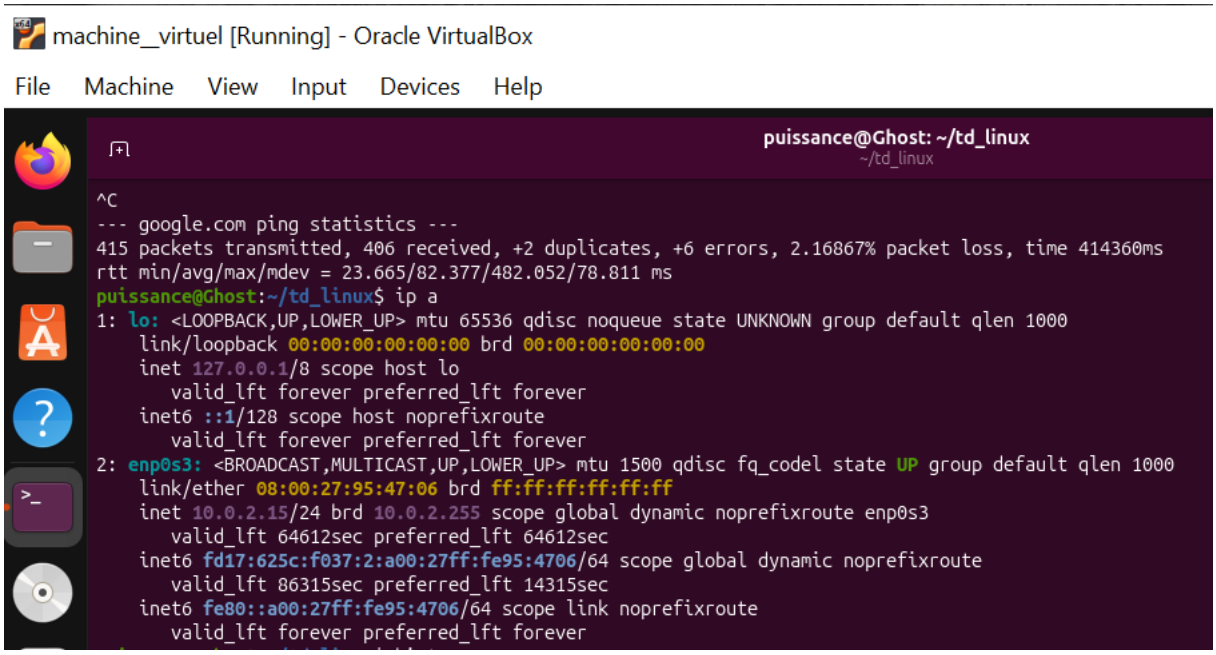


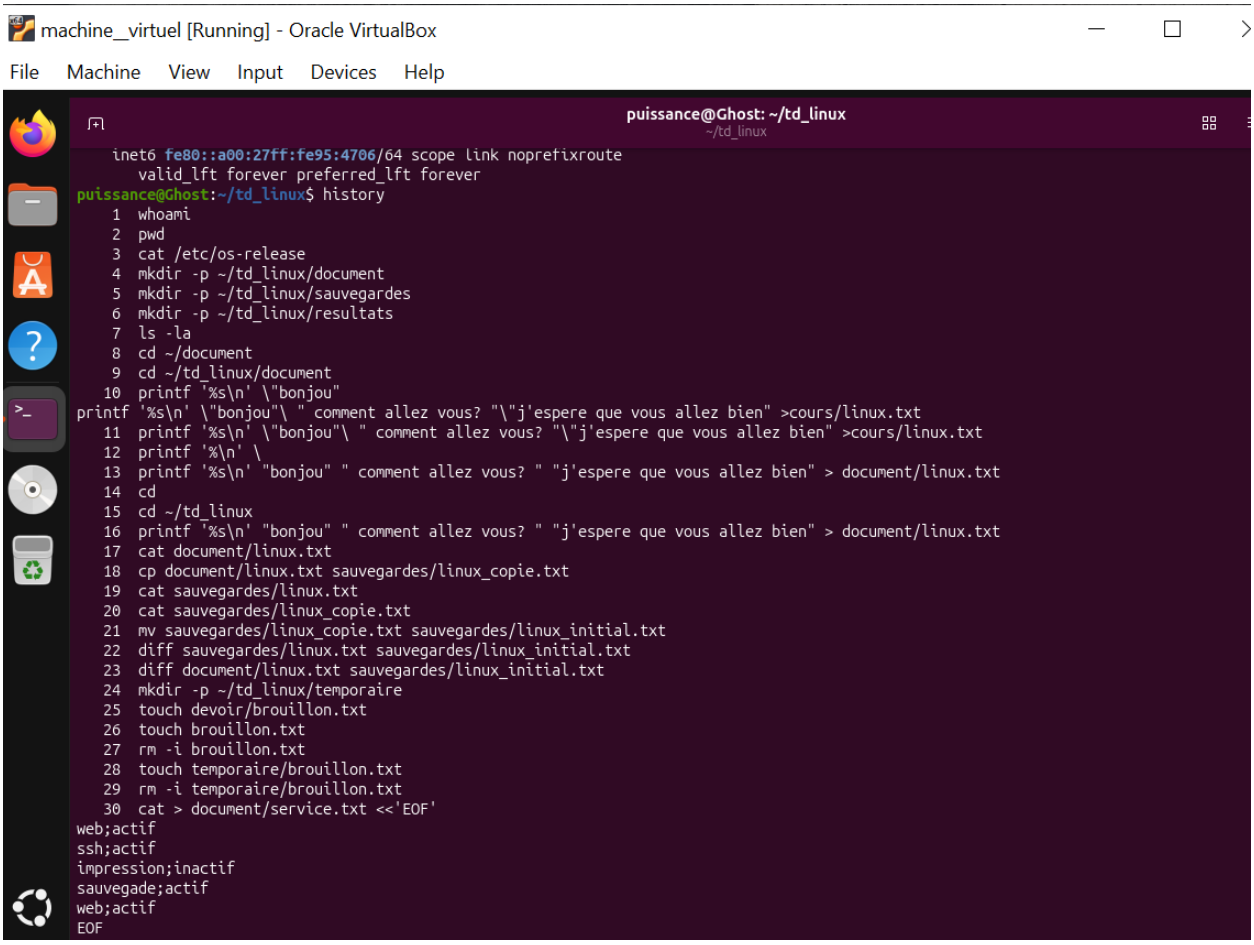
7. Simuler la commande autoremove

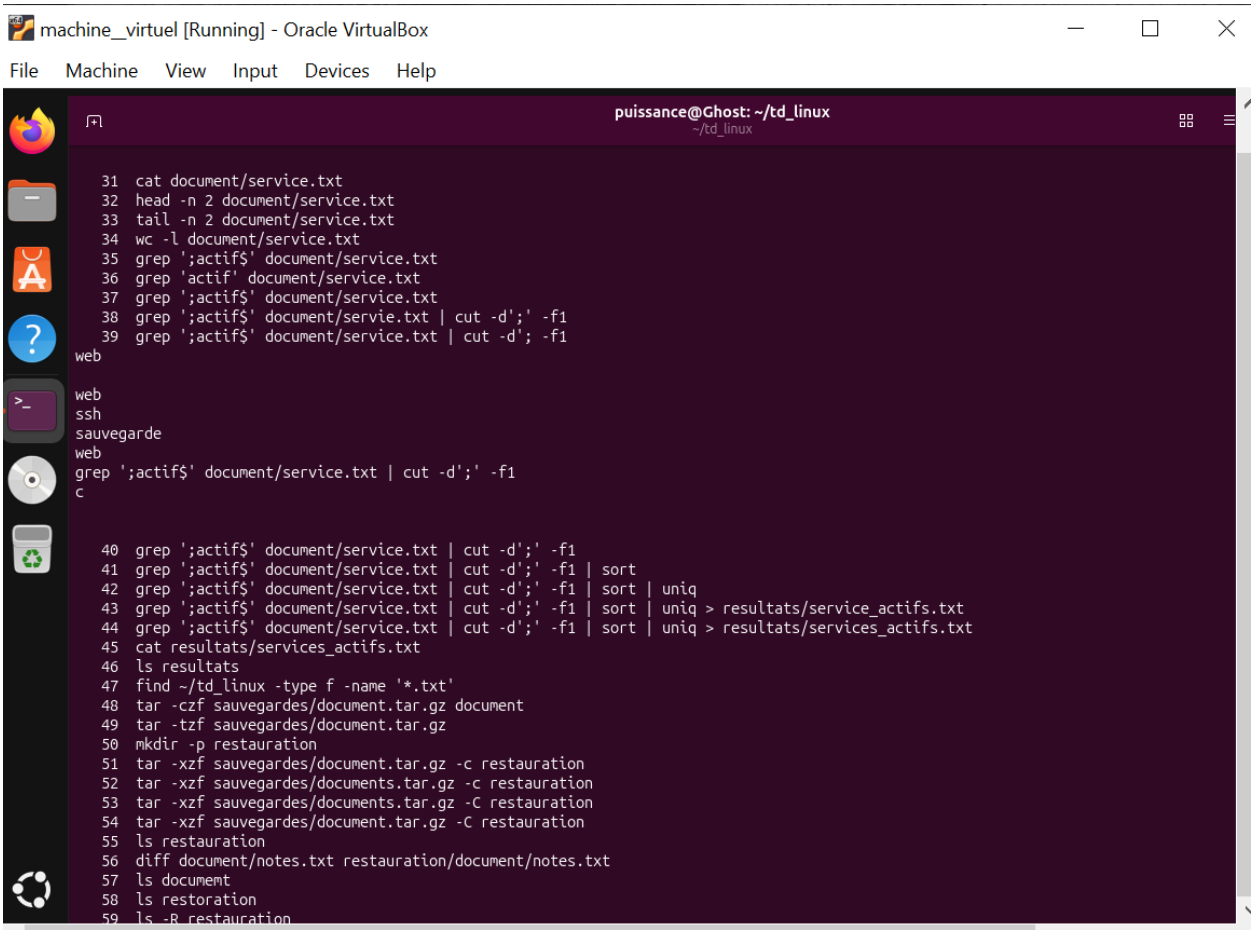

machine_virtuel [Running] - Oracle VirtualBox
File Machine View Input Devices Help
puissance@Ghost: ~/td_linux
~/td_linux
Suggests: libnl-3-200, libnl-genl-3-200, lm-sensors, lsof, strace
Homepage: https://htop.dev/
Task: cloud-image, cloud-image, server, ubuntu-server-raspi, kubuntu-desktop, kubuntu-full, lubuntu-desktop, ubuntu-mate-core, ubuntu-
Download-Size: 235 kB
APT-Sources: http://ht.archive.ubuntu.com/ubuntu stonking/main amd64 Packages
Description: interactive processes viewer
Htop is an ncursed-based process viewer similar to top, but it
allows one to scroll the list vertically and horizontally to see
all processes and their full command lines.
.
Tasks related to processes (killing, renicing) can be done without
entering their PIDs.
puissance@Ghost:~/td_linux$ apt -s install htop
NOTE: This is only a simulation!
apt needs root privileges for real execution.
Keep also in mind that locking is deactivated,
so don't depend on the relevance to the real current situation!
Installing:
htop
Suggested packages:
lm-sensors
Summary:
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 15
Inst htop (3.5.3-1 Ubuntu:26.10/stonking [amd64])
Conf htop (3.5.3-1 Ubuntu:26.10/stonking [amd64])
puissance@Ghost:~/td_linux$ apt -s autoremove
NOTE: This is only a simulation!
apt needs root privileges for real execution.
Keep also in mind that locking is deactivated,
so don't depend on the relevance to the real current situation!
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 15
puissance@Ghost:~/td_linux$

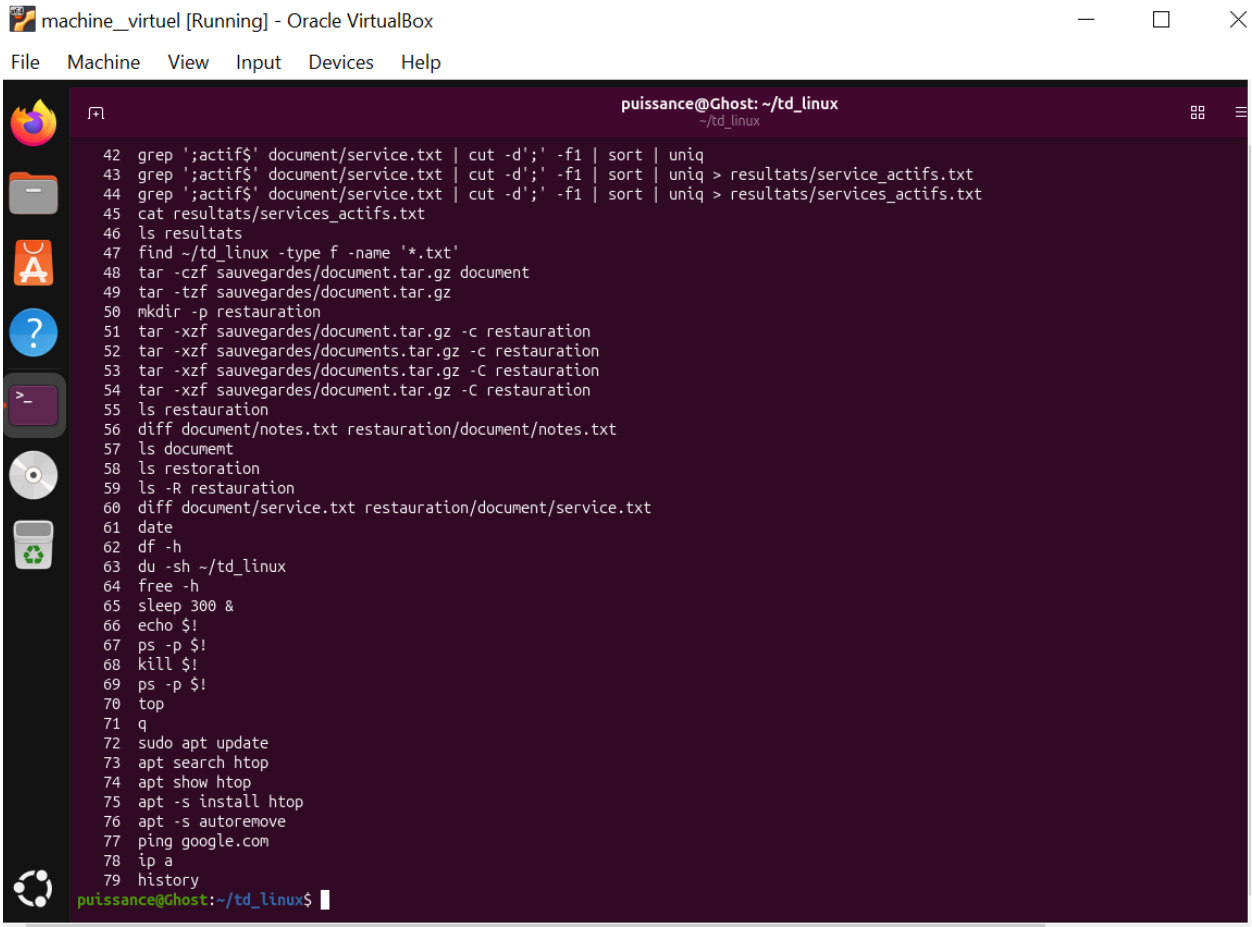
Bloc « Exécutez ces commandes »



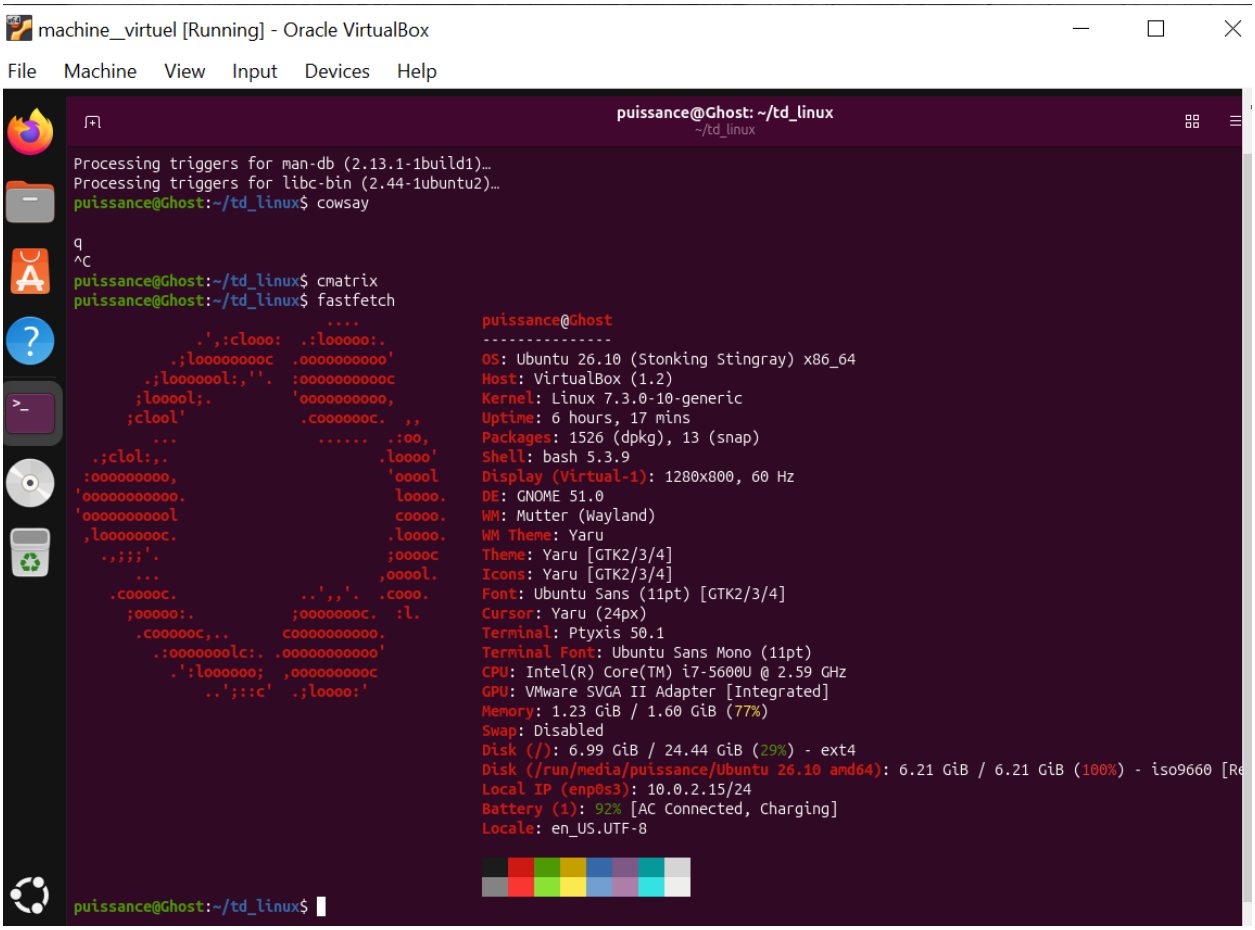


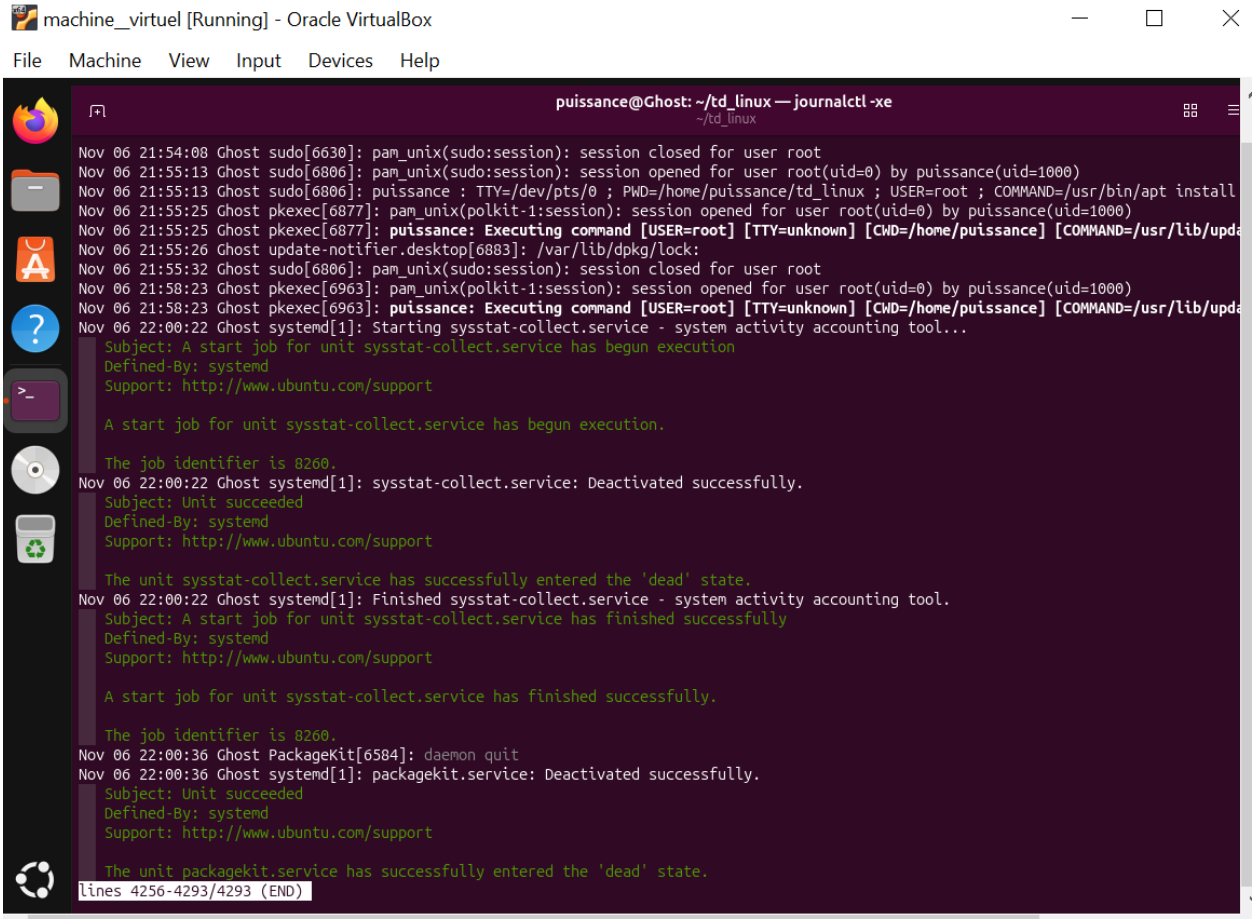

machine_virtuel [Running] - Oracle VirtualBox
File Machine View Input Devices Help
puissance@Ghost: ~/td_linux
~/td_linux
inet6 fe80::a00:27ff:fe95:4706/64 scope link noprefixroute
valid_lft forever preferred_lft forever
puissance@Ghost:~/td_linux$ history
1 whoami
2 pwd
3 cat /etc/os-release
4 mkdir -p ~/td_linux/document
5 mkdir -p ~/td_linux/sauvegardes
6 mkdir -p ~/td_linux/resultats
7 ls -la
8 cd ~/document
9 cd ~/td_linux/document
10 printf '%s\n' \"bonjou"
printf '%s\n' \"bonjou"\ " comment allez vous? "\"j'espere que vous allez bien" >cours/linux.txt
11 printf '%s\n' \"bonjou"\ " comment allez vous? "\"j'espere que vous allez bien" >cours/linux.txt
12 printf '%\n' \
13 printf '%s\n' "bonjou" " comment allez vous? " "j'espere que vous allez bien" > document/linux.txt
14 cd
15 cd ~/td_linux
16 printf '%s\n' "bonjou" " comment allez vous? " "j'espere que vous allez bien" > document/linux.txt
17 cat document/linux.txt
18 cp document/linux.txt sauvegardes/linux_copie.txt
19 cat sauvegardes/linux.txt
20 cat sauvegardes/linux_copie.txt
21 mv sauvegardes/linux_copie.txt sauvegardes/linux_initial.txt
22 diff sauvegardes/linux.txt sauvegardes/linux_initial.txt
23 diff document/linux.txt sauvegardes/linux_initial.txt
24 mkdir -p ~/td_linux/temporaire
25 touch devoir/brouillon.txt
26 touch brouillon.txt
27 rm -i brouillon.txt
28 touch temporaire/brouillon.txt
29 rm -i temporaire/brouillon.txt
30 cat > document/service.txt <<'EOF'
web;actif
ssh;actif
impression;inactif
sauvegade;actif
web;actif
EOF

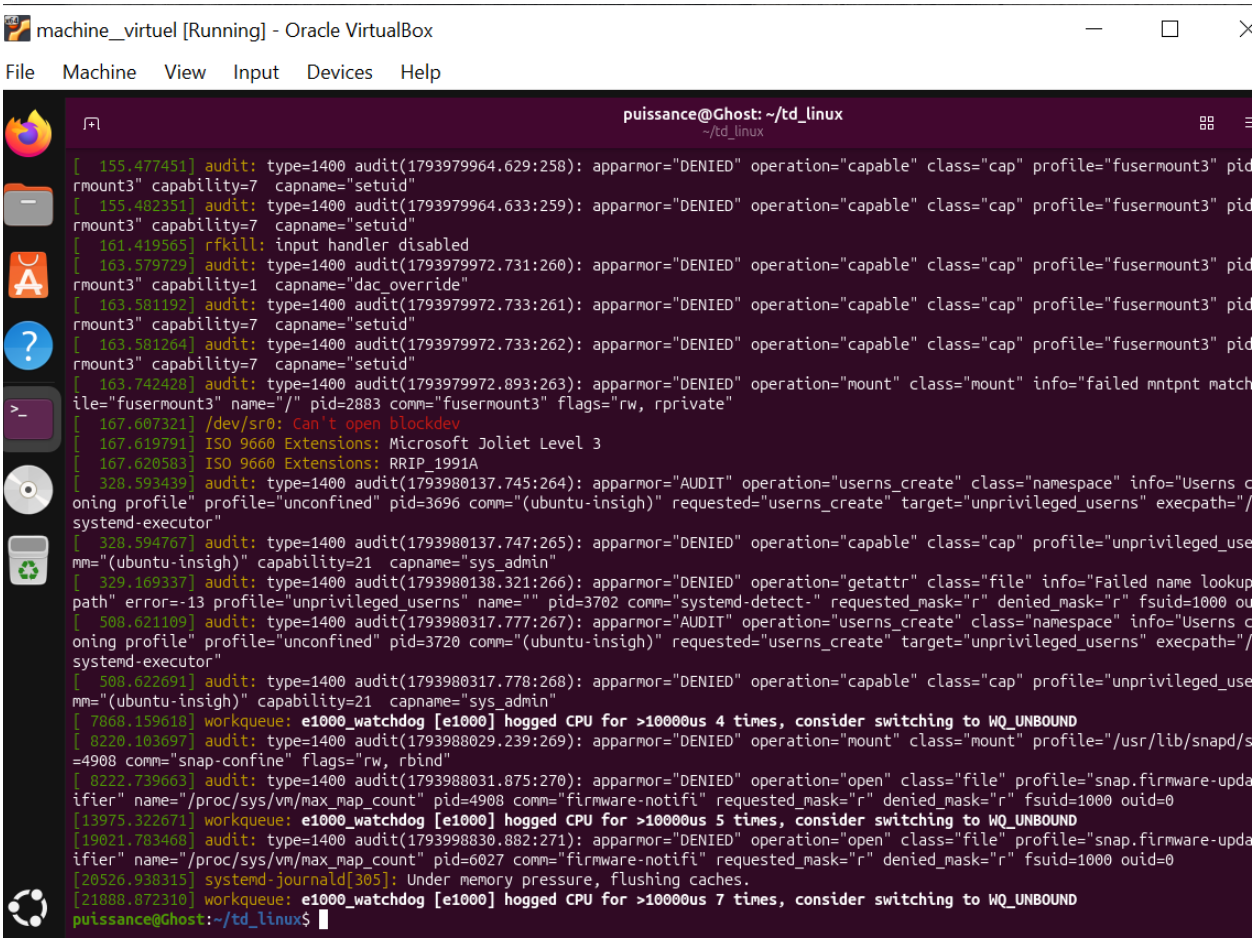

machine_virtuel [Running] - Oracle VirtualBox
File Machine View Input Devices Help
puissance@Ghost: ~/td_linux
~/td_linux
31 cat document/service.txt
32 head -n 2 document/service.txt
33 tail -n 2 document/service.txt
34 wc -l document/service.txt
35 grep ';actif$' document/service.txt
36 grep 'actif' document/service.txt
37 grep ';actif$' document/service.txt
38 grep ';actif$' document/servie.txt | cut -d';' -f1
39 grep ';actif$' document/service.txt | cut -d'; -f1
web
web
ssh
sauvegarde
web
grep ';actif$' document/service.txt | cut -d';' -f1
c
40 grep ';actif$' document/service.txt | cut -d';' -f1
41 grep ';actif$' document/service.txt | cut -d';' -f1 | sort
42 grep ';actif$' document/service.txt | cut -d';' -f1 | sort | uniq
43 grep ';actif$' document/service.txt | cut -d';' -f1 | sort | uniq > resultats/service_actifs.txt
44 grep ';actif$' document/service.txt | cut -d';' -f1 | sort | uniq > resultats/services_actifs.txt
45 cat resultats/services_actifs.txt
46 ls resultats
47 find ~/td_linux -type f -name '*.txt'
48 tar -czf sauvegardes/document.tar.gz document
49 tar -tzf sauvegardes/document.tar.gz
50 mkdir -p restauration
51 tar -xzf sauvegardes/document.tar.gz -c restauration
52 tar -xzf sauvegardes/documents.tar.gz -c restauration
53 tar -xzf sauvegardes/documents.tar.gz -C restauration
54 tar -xzf sauvegardes/document.tar.gz -C restauration
55 ls restauration
56 diff document/notes.txt restauration/document/notes.txt
57 ls documemt
58 ls restoration
59 ls -R restauration

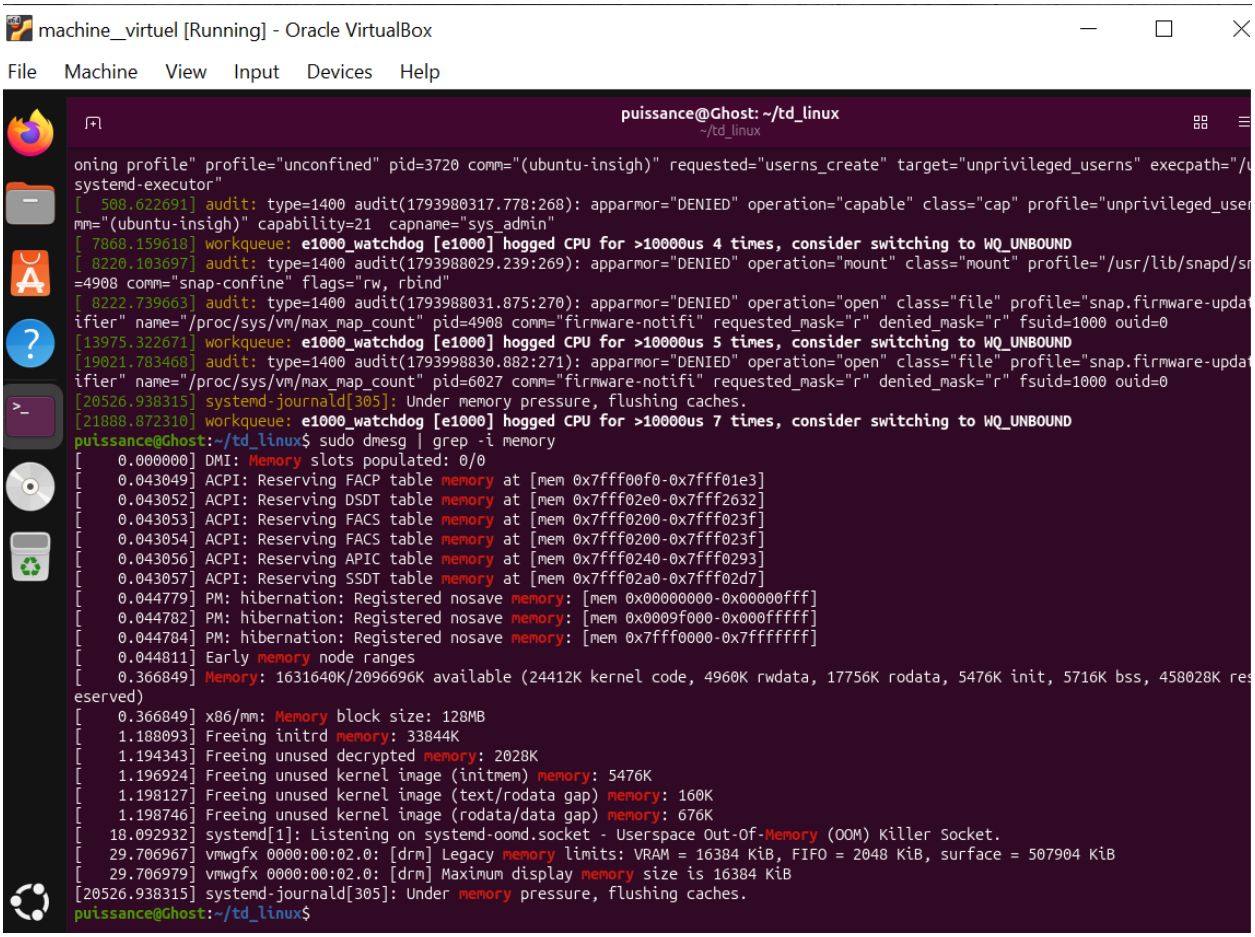

machine_virtuel [Running] - Oracle VirtualBox
File Machine View Input Devices Help
puissance@Ghost: ~/td_linux
~/td_linux
42 grep ';actif$' document/service.txt | cut -d';' -f1 | sort | uniq
43 grep ';actif$' document/service.txt | cut -d';' -f1 | sort | uniq > resultats/service_actifs.txt
44 grep ';actif$' document/service.txt | cut -d';' -f1 | sort | uniq > resultats/services_actifs.txt
45 cat resultats/services_actifs.txt
46 ls resultats
47 find ~/td_linux -type f -name '*.txt'
48 tar -czf sauvegardes/document.tar.gz document
49 tar -tzf sauvegardes/document.tar.gz
50 mkdir -p restauration
51 tar -xzf sauvegardes/document.tar.gz -c restauration
52 tar -xzf sauvegardes/documents.tar.gz -c restauration
53 tar -xzf sauvegardes/documents.tar.gz -C restauration
54 tar -xzf sauvegardes/document.tar.gz -C restauration
55 ls restauration
56 diff document/notes.txt restauration/document/notes.txt
57 ls documemt
58 ls restoration
59 ls -R restauration
60 diff document/service.txt restauration/document/service.txt
61 date
62 df -h
63 du -sh ~/td_linux
64 free -h
65 sleep 300 &
66 echo $!
67 ps -p $!
68 kill $!
69 ps -p $!
70 top
71 q
72 sudo apt update
73 apt search htop
74 apt show htop
75 apt -s install htop
76 apt -s autoremove
77 ping google.com
78 ip a
79 history
puissance@Ghost:~/td_linux$

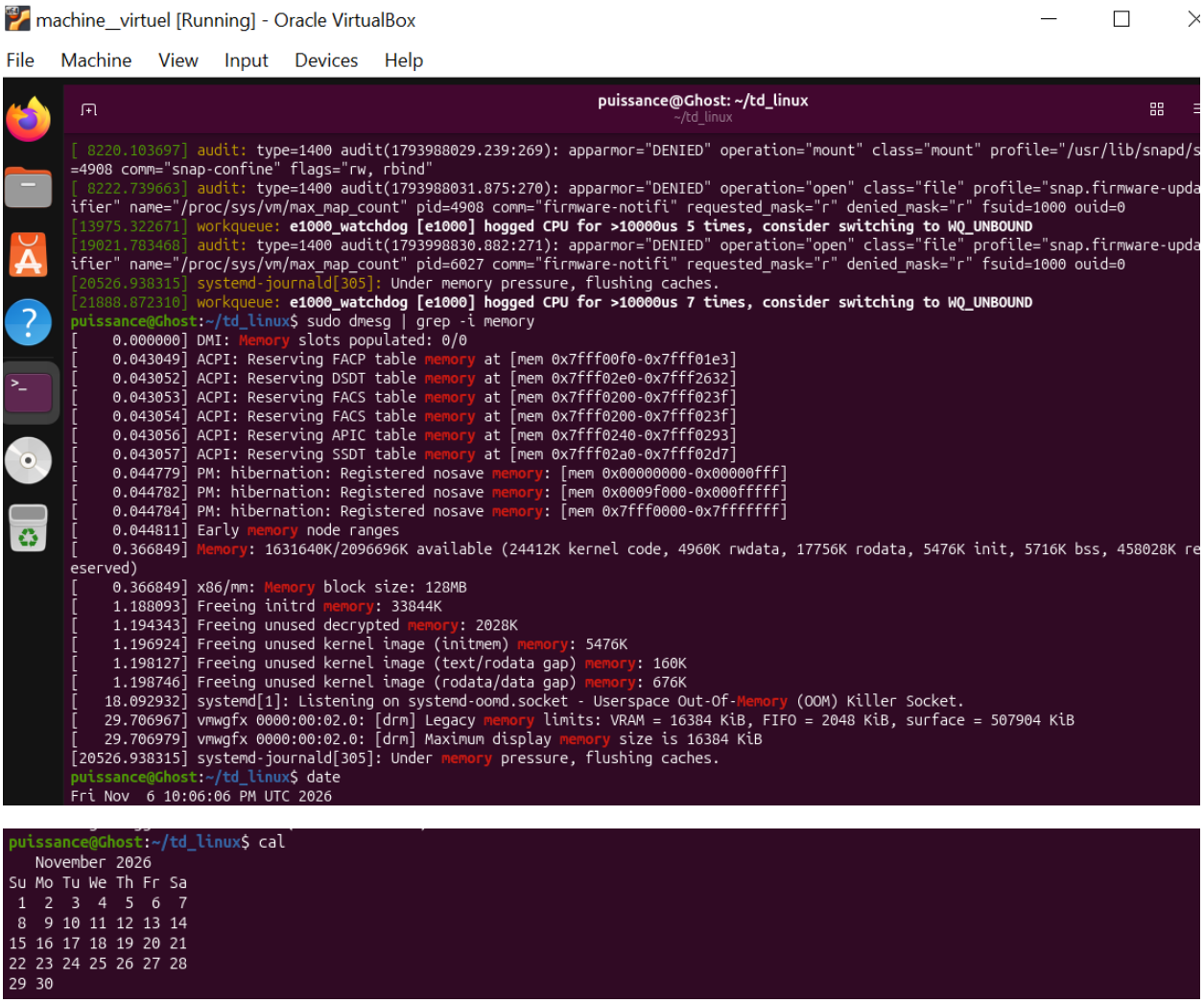

machine_virtuel [Running] - Oracle VirtualBox
File Machine View Input Devices Help
puissance@Ghost: ~/td_linux — cmatrix
~/td_linux


machine_virtuel [Running] - Oracle VirtualBox
File Machine View Input Devices Help
puissance@Ghost: ~/td_linux
~/td_linux
Processing triggers for man-db (2.13.1-1build1)…
Processing triggers for libc-bin (2.44-1ubuntu2)…
puissance@Ghost:~/td_linux$ cowsay
q
^C
puissance@Ghost:~/td_linux$ cmatrix
puissance@Ghost:~/td_linux$ fastfetch
puissance@Ghost
OS: Ubuntu 26.10 (Stonking Stingray) x86_64
Host: VirtualBox (1.2)
Kernel: Linux 7.3.0-10-generic
Uptime: 6 hours, 17 mins
Packages: 1526 (dpkg), 13 (snap)
Shell: bash 5.3.9
Display (Virtual-1): 1280x800, 60 Hz
DE: GNOME 51.0
WM: Mutter (Wayland)
WM Theme: Yaru
Theme: Yaru [GTK2/3/4]
Icons: Yaru [GTK2/3/4]
Font: Ubuntu Sans (11pt) [GTK2/3/4]
Cursor: Yaru (24px)
Terminal: Ptyxis 50.1
Terminal Font: Ubuntu Sans Mono (11pt)
CPU: Intel(R) Core(TM) i7-5600U @ 2.59 GHz
GPU: VMware SVGA II Adapter [Integrated]
Memory: 1.23 GiB / 1.60 GiB (77%)
Swap: Disabled
Disk (/): 6.99 GiB / 24.44 GiB (29%) - ext4
Disk (/run/media/puissance/Ubuntu 26.10 amd64): 6.21 GiB / 6.21 GiB (100%) - iso9660 [Re
Local IP (enp0s3): 10.0.2.15/24
Battery (1): 92% [AC Connected, Charging]
Locale: en_US.UTF-8
puissance@Ghost:~/td_linux$


machine_virtuel [Running] - Oracle VirtualBox
File Machine View Input Devices Help
puissance@Ghost: ~/td_linux — journalctl -xe
~/td_linux
Nov 06 21:54:08 Ghost sudo[6630]: pam_unix(sudo:session): session closed for user root
Nov 06 21:55:13 Ghost sudo[6806]: pam_unix(sudo:session): session opened for user root(uid=0) by puissance(uid=1000)
Nov 06 21:55:13 Ghost sudo[6806]: puissance : TTY=/dev/pts/0 ; PWD=/home/puissance/td_linux ; USER=root ; COMMAND=/usr/bin/apt install
Nov 06 21:55:25 Ghost pkexec[6877]: pam_unix(polkit-1:session): session opened for user root(uid=0) by puissance(uid=1000)
Nov 06 21:55:25 Ghost pkexec[6877]: puissance: Executing command [USER=root] [TTY=unknown] [CWD=/home/puissance] [COMMAND=/usr/lib/upda
Nov 06 21:55:26 Ghost update-notifier.desktop[6883]: /var/lib/dpkg/lock:
Nov 06 21:55:32 Ghost sudo[6806]: pam_unix(sudo:session): session closed for user root
Nov 06 21:58:23 Ghost pkexec[6963]: pam_unix(polkit-1:session): session opened for user root(uid=0) by puissance(uid=1000)
Nov 06 21:58:23 Ghost pkexec[6963]: puissance: Executing command [USER=root] [TTY=unknown] [CWD=/home/puissance] [COMMAND=/usr/lib/upda
Nov 06 22:00:22 Ghost systemd[1]: Starting sysstat-collect.service - system activity accounting tool...
Subject: A start job for unit sysstat-collect.service has begun execution
Defined-By: systemd
Support: http://www.ubuntu.com/support
A start job for unit sysstat-collect.service has begun execution.
The job identifier is 8260.
Nov 06 22:00:22 Ghost systemd[1]: sysstat-collect.service: Deactivated successfully.
Subject: Unit succeeded
Defined-By: systemd
Support: http://www.ubuntu.com/support
The unit sysstat-collect.service has successfully entered the 'dead' state.
Nov 06 22:00:22 Ghost systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
Subject: A start job for unit sysstat-collect.service has finished successfully
Defined-By: systemd
Support: http://www.ubuntu.com/support
A start job for unit sysstat-collect.service has finished successfully.
The job identifier is 8260.
Nov 06 22:00:36 Ghost PackageKit[6584]: daemon quit
Nov 06 22:00:36 Ghost systemd[1]: packagekit.service: Deactivated successfully.
Subject: Unit succeeded
Defined-By: systemd
Support: http://www.ubuntu.com/support
The unit packagekit.service has successfully entered the 'dead' state.
lines 4256-4293/4293 (END)


machine_virtuel [Running] - Oracle VirtualBox
File Machine View Input Devices Help
puissance@Ghost: ~/td_linux
~/td_linux
[155.477451] audit: type=1400 audit(1793979964.629:258): apparmor="DENIED" operation="capable" class="cap" profile="fusermount3" pid
rmount3" capability=7 capname="setuid"
[155.482351] audit: type=1400 audit(1793979964.633:259): apparmor="DENIED" operation="capable" class="cap" profile="fusermount3" pid
rmount3" capability=7 capname="setuid"
[161.419565] rfkill: input handler disabled
[163.579729] audit: type=1400 audit(1793979972.731:260): apparmor="DENIED" operation="capable" class="cap" profile="fusermount3" pid
rmount3" capability=1 capname="dac_override"
[163.581192] audit: type=1400 audit(1793979972.733:261): apparmor="DENIED" operation="capable" class="cap" profile="fusermount3" pid
rmount3" capability=7 capname="setuid"
[163.581264] audit: type=1400 audit(1793979972.733:262): apparmor="DENIED" operation="capable" class="cap" profile="fusermount3" pid
rmount3" capability=7 capname="setuid"
[163.742428] audit: type=1400 audit(1793979972.893:263): apparmor="DENIED" operation="mount" class="mount" info="failed mntpnt match
ile="fusermount3" name="/" pid=2883 comm="fusermount3" flags="rw, rprivate"
[167.607321] /dev/sr0: Can't open blockdev
[167.619791] ISO 9660 Extensions: Microsoft Joliet Level 3
[167.620583] ISO 9660 Extensions: RRIP_1991A
[328.593439] audit: type=1400 audit(1793980137.745:264): apparmor="AUDIT" operation="userns_create" class="namespace" info="Userns c
oning profile" profile="unconfined" pid=3696 comm="(ubuntu-insigh)" requested="userns_create" target="unprivileged_userns" execpath="/
systemd-executor"
[328.594767] audit: type=1400 audit(1793980137.747:265): apparmor="DENIED" operation="capable" class="cap" profile="unprivileged_use
mm="(ubuntu-insigh)" capability=21 capname="sys_admin"
[329.169337] audit: type=1400 audit(1793980138.321:266): apparmor="DENIED" operation="getattr" class="file" info="Failed name lookup
path" error=-13 profile="unprivileged_userns" name="" pid=3702 comm="systemd-detect-" requested_mask="r" denied_mask="r" fsuid=1000 ou
[508.621109] audit: type=1400 audit(1793980317.777:267): apparmor="AUDIT" operation="userns_create" class="namespace" info="Userns c
oning profile" profile="unconfined" pid=3720 comm="(ubuntu-insigh)" requested="userns_create" target="unprivileged_userns" execpath="/
systemd-executor"
[508.622691] audit: type=1400 audit(1793980317.778:268): apparmor="DENIED" operation="capable" class="cap" profile="unprivileged_use
mm="(ubuntu-insigh)" capability=21 capname="sys_admin"
[7868.159618] workqueue: e1000_watchdog [e1000] hogged CPU for >10000us 4 times, consider switching to WQ_UNBOUND
[8220.103697] audit: type=1400 audit(1793988029.239:269): apparmor="DENIED" operation="mount" class="mount" profile="/usr/lib/snapd/s
=4908 comm="snap-confine" flags="rw, rbind"
[8222.739663] audit: type=1400 audit(1793988031.875:270): apparmor="DENIED" operation="open" class="file" profile="snap.firmware-upda
ifier" name="/proc/sys/vm/max_map_count" pid=4908 comm="firmware-notifi" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[13975.322671] workqueue: e1000_watchdog [e1000] hogged CPU for >10000us 5 times, consider switching to WQ_UNBOUND
[19021.783468] audit: type=1400 audit(1793998830.882:271): apparmor="DENIED" operation="open" class="file" profile="snap.firmware-upda
ifier" name="/proc/sys/vm/max_map_count" pid=6027 comm="firmware-notifi" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[20526.938315] systemd-journald[305]: Under memory pressure, flushing caches.
[21888.872310] workqueue: e1000_watchdog [e1000] hogged CPU for >10000us 7 times, consider switching to WQ_UNBOUND
puissance@Ghost:~/td_linux$


machine_virtuel [Running] - Oracle VirtualBox
File Machine View Input Devices Help
puissance@Ghost: ~/td_linux
~/td_linux
oning profile" profile="unconfined" pid=3720 comm="(ubuntu-insigh)" requested="userns_create" target="unprivileged_userns" execpath="/
systemd-executor"
[508.622691] audit: type=1400 audit(1793980317.778:268): apparmor="DENIED" operation="capable" class="cap" profile="unprivileged_user
mm="(ubuntu-insigh)" capability=21 capname="sys_admin"
[7868.159618] workqueue: e1000_watchdog [e1000] hogged CPU for >10000us 4 times, consider switching to WQ_UNBOUND
[8220.103697] audit: type=1400 audit(1793988029.239:269): apparmor="DENIED" operation="mount" class="mount" profile="/usr/lib/snapd/sn
=4908 comm="snap-confine" flags="rw, rbind"
[8222.739663] audit: type=1400 audit(1793988031.875:270): apparmor="DENIED" operation="open" class="file" profile="snap.firmware-updat
ifier" name="/proc/sys/vm/max_map_count" pid=4908 comm="firmware-notifi" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[13975.322671] workqueue: e1000_watchdog [e1000] hogged CPU for >10000us 5 times, consider switching to WQ_UNBOUND
[19021.783468] audit: type=1400 audit(1793998830.882:271): apparmor="DENIED" operation="open" class="file" profile="snap.firmware-updat
ifier" name="/proc/sys/vm/max_map_count" pid=6027 comm="firmware-notifi" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[20526.938315] systemd-journald[305]: Under memory pressure, flushing caches.
[21888.872310] workqueue: e1000_watchdog [e1000] hogged CPU for >10000us 7 times, consider switching to WQ_UNBOUND
puissance@Ghost:~/td_linux$ sudo dmesg | grep -i memory
[0.000000] DMI: Memory slots populated: 0/0
[0.043049] ACPI: Reserving FACP table memory at [mem 0x7fff00f0-0x7fff01e3]
[0.043052] ACPI: Reserving DSDT table memory at [mem 0x7fff02e0-0x7fff2632]
[0.043053] ACPI: Reserving FACS table memory at [mem 0x7fff0200-0x7fff023f]
[0.043054] ACPI: Reserving FACS table memory at [mem 0x7fff0200-0x7fff023f]
[0.043056] ACPI: Reserving APIC table memory at [mem 0x7fff0240-0x7fff0293]
[0.043057] ACPI: Reserving SSDT table memory at [mem 0x7fff02a0-0x7fff02d7]
[0.044779] PM: hibernation: Registered nosave memory: [mem 0x00000000-0x00000fff]
[0.044782] PM: hibernation: Registered nosave memory: [mem 0x0009f000-0x000fffff]
[0.044784] PM: hibernation: Registered nosave memory: [mem 0x7fff0000-0x7fffffff]
[0.044811] Early memory node ranges
[0.366849] Memory: 1631640K/2096696K available (24412K kernel code, 4960K rwdata, 17756K rodata, 5476K init, 5716K bss, 458028K res
eserved)
[0.366849] x86/mm: Memory block size: 128MB
[1.188093] Freeing initrd memory: 33844K
[1.194343] Freeing unused decrypted memory: 2028K
[1.196924] Freeing unused kernel image (initmem) memory: 5476K
[1.198127] Freeing unused kernel image (text/rodata gap) memory: 160K
[1.198746] Freeing unused kernel image (rodata/data gap) memory: 676K
[18.092932] systemd[1]: Listening on systemd-oomd.socket - Userspace Out-Of-Memory (OOM) Killer Socket.
[29.706967] vmwgfx 0000:00:02.0: [drm] Legacy memory limits: VRAM = 16384 KiB, FIFO = 2048 KiB, surface = 507904 KiB
[29.706979] vmwgfx 0000:00:02.0: [drm] Maximum display memory size is 16384 KiB
[20526.938315] systemd-journald[305]: Under memory pressure, flushing caches.
puissance@Ghost:~/td_linux$


machine_virtuel [Running] - Oracle VirtualBox
File Machine View Input Devices Help
puissance@Ghost: ~/td_linux
~/td_linux
[8220.103697] audit: type=1400 audit(1793988029.239:269): apparmor="DENIED" operation="mount" class="mount" profile="/usr/lib/snapd/s
=4908 comm="snap-confine" flags="rw, rbind"
[8222.739663] audit: type=1400 audit(1793988031.875:270): apparmor="DENIED" operation="open" class="file" profile="snap.firmware-upda
ifier" name="/proc/sys/vm/max_map_count" pid=4908 comm="firmware-notifi" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[13975.322671] workqueue: e1000_watchdog [e1000] hogged CPU for >10000us 5 times, consider switching to WQ_UNBOUND
[19021.783468] audit: type=1400 audit(1793998830.882:271): apparmor="DENIED" operation="open" class="file" profile="snap.firmware-upda
ifier" name="/proc/sys/vm/max_map_count" pid=6027 comm="firmware-notifi" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[20526.938315] systemd-journald[305]: Under memory pressure, flushing caches.
[21888.872310] workqueue: e1000_watchdog [e1000] hogged CPU for >10000us 7 times, consider switching to WQ_UNBOUND
puissance@Ghost:~/td_linux$ sudo dmesg | grep -i memory
[0.000000] DMI: Memory slots populated: 0/0
[0.043049] ACPI: Reserving FACP table memory at [mem 0x7fff00f0-0x7fff01e3]
[0.043052] ACPI: Reserving DSDT table memory at [mem 0x7fff02e0-0x7fff2632]
[0.043053] ACPI: Reserving FACS table memory at [mem 0x7fff0200-0x7fff023f]
[0.043054] ACPI: Reserving FACS table memory at [mem 0x7fff0200-0x7fff023f]
[0.043056] ACPI: Reserving APIC table memory at [mem 0x7fff0240-0x7fff0293]
[0.043057] ACPI: Reserving SSDT table memory at [mem 0x7fff02a0-0x7fff02d7]
[0.044779] PM: hibernation: Registered nosave memory: [mem 0x00000000-0x00000fff]
[0.044782] PM: hibernation: Registered nosave memory: [mem 0x0009f000-0x000fffff]
[0.044784] PM: hibernation: Registered nosave memory: [mem 0x7fff0000-0x7fffffff]
[0.044811] Early memory node ranges
[0.366849] Memory: 1631640K/2096696K available (24412K kernel code, 4960K rwdata, 17756K rodata, 5476K init, 5716K bss, 458028K re
eserved)
[0.366849] x86/mm: Memory block size: 128MB
[1.188093] Freeing initrd memory: 33844K
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[1.196924] Freeing unused kernel image (initmem) memory: 5476K
[1.198127] Freeing unused kernel image (text/rodata gap) memory: 160K
[1.198746] Freeing unused kernel image (rodata/data gap) memory: 676K
[18.092932] systemd[1]: Listening on systemd-oomd.socket - Userspace Out-Of-Memory (OOM) Killer Socket.
[29.706967] vmwgfx 0000:00:02.0: [drm] Legacy memory limits: VRAM = 16384 KiB, FIFO = 2048 KiB, surface = 507904 KiB
[29.706979] vmwgfx 0000:00:02.0: [drm] Maximum display memory size is 16384 KiB
[20526.938315] systemd-journald[305]: Under memory pressure, flushing caches.
puissance@Ghost:~/td_linux$ date
Fri Nov 6 10:06:06 PM UTC 2026

puissance@Ghost:~/td_linux$ cal
November 2026
Su Mo Tu We Th Fr Sa
1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30

```
puissance@Ghost:~/td_linux$ sudo apt install toilet
Installing:
  toilet

Installing dependencies:
  toilet-fonts

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 15
  Download size: 726 kB
  Space needed: 849 kB / 17.4 GB available

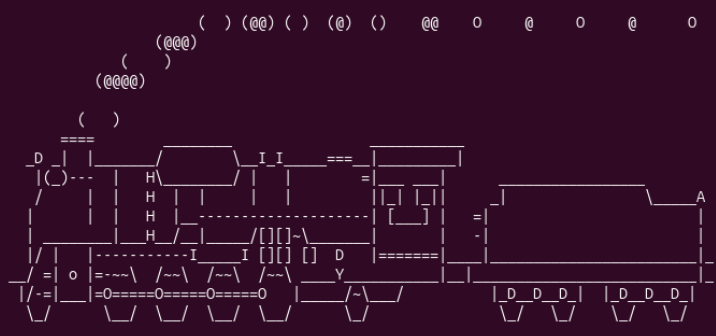
Continue? [Y/n] y
Get:1 http://ht.archive.ubuntu.com/ubuntu stonking/universe amd64 toilet-fonts all 0.3-6 [713 kB]
Get:2 http://ht.archive.ubuntu.com/ubuntu stonking/universe amd64 toilet amd64 0.3-6 [13.0 kB]
Fetched 726 kB in 2s (358 kB/s)
Selecting previously unselected package toilet-fonts.
(Reading database… 197701 files and directories currently installed.)
Preparing to unpack …/toilet-fonts_0.3-6_all.deb…
Unpacking toilet-fonts (0.3-6)…
Selecting previously unselected package toilet.
Preparing to unpack …/toilet_0.3-6_amd64.deb…
Unpacking toilet (0.3-6)…
Setting up toilet-fonts (0.3-6)…
Setting up toilet (0.3-6)…
Processing triggers for man-db (2.13.1-1build1)…
puissance@Ghost:~/td_linux$ figlet "UBUNTU"
 _   _ ____  _   _ _____ _   _
| | | | __ )| | | |_   _| | | |
| | | |  _ \| | | | | | | | | |
| |_| | |_) | |_| | | | | |_| |
 \___/|____/ \___/|_| |_|  \___/

puissance@Ghost:~/td_linux$
```

machine_virtuel [Running] - Oracle VirtualBox

File Machine View Input Devices Help

puissance@Ghost: ~/td_linux — sl
~/td_linux



```
puissance@Ghost:~/td_linux$ seq 1 10
1
2
3
4
5
6
7
8
9
10
puissance@Ghost:~/td_linux$
```

```
puissance@Ghost:~/td_linux$ seq 2 2 20
2
4
6
8
10
12
14
16
18
20
puissance@Ghost:~/td_linux$
```

machine_virtuel [Running] - Oracle VirtualBox

File Machine View Input Devices Help

puissance@Ghost: ~/td_linux
~/td_linux

```
  Download size: 12.9 kB
  Space needed: 60.4 kB / 17.4 GB available

Get:1 http://ht.archive.ubuntu.com/ubuntu stonking/universe amd64 sl amd64 5.02-1build1 [12.9 kB]
Fetched 12.9 kB in 1s (15.1 kB/s)
Selecting previously unselected package sl.
(Reading database… 197738 files and directories currently installed.)
Preparing to unpack …/sl_5.02-1build1_amd64.deb…
Unpacking sl (5.02-1build1)…
Setting up sl (5.02-1build1)…
Processing triggers for man-db (2.13.1-1build1)…
puissance@Ghost:~/td_linux$ sl

puissance@Ghost:~/td_linux$ sl
puissance@Ghost:~/td_linux$ seq 1 10
1
2
3
4
5
6
7
8
9
10
puissance@Ghost:~/td_linux$ seq 2 2 20
2
4
6
8
10
12
14
16
18
20
puissance@Ghost:~/td_linux$ shuf -i 1-100 -n 1
67
puissance@Ghost:~/td_linux$
```

EXERCICE 5 – KDE Plasma et XFCE

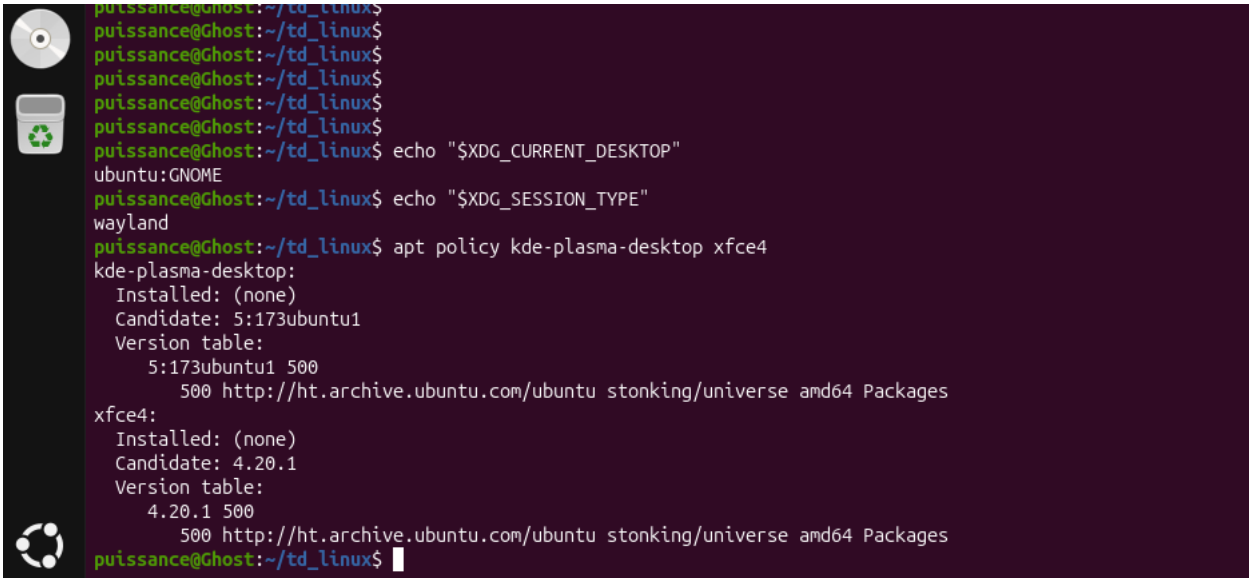
Un environnement de bureau fournit une interface graphique et des outils. Le gestionnaire de connexion présente l'écran d'ouverture de session et permet généralement de choisir le bureau.

1. Bureau et type de session actuels

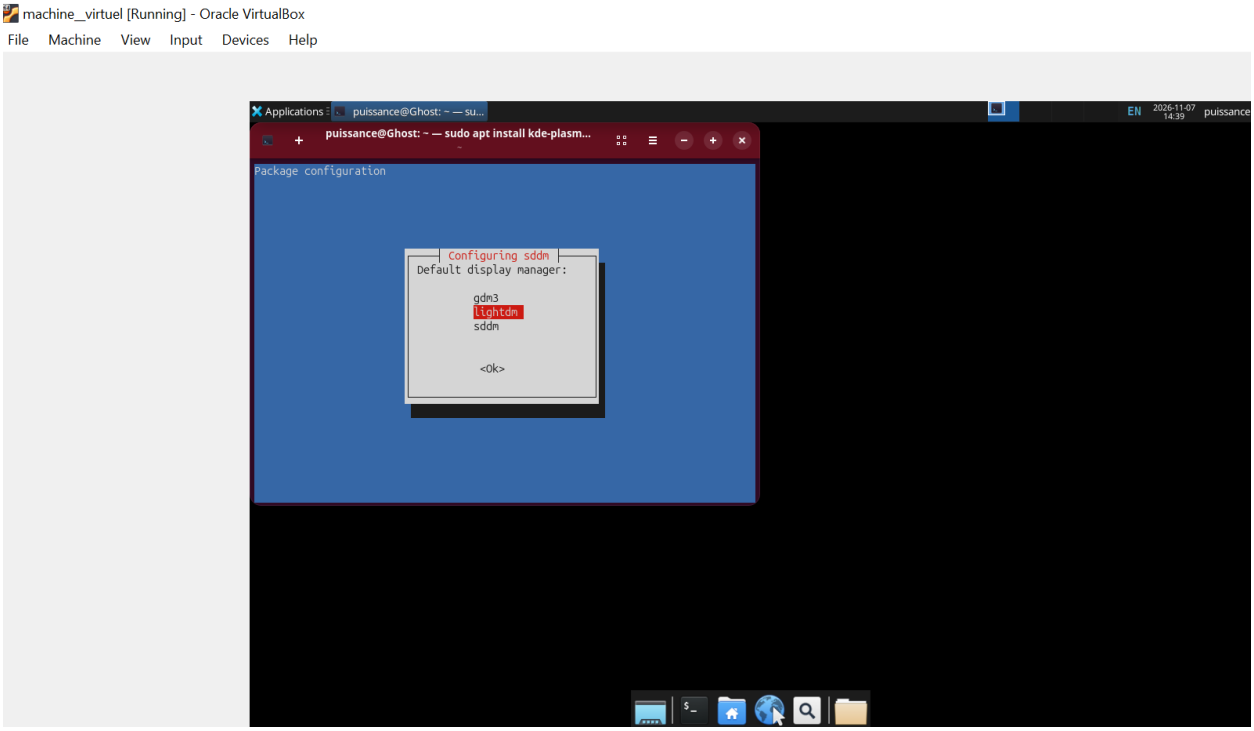


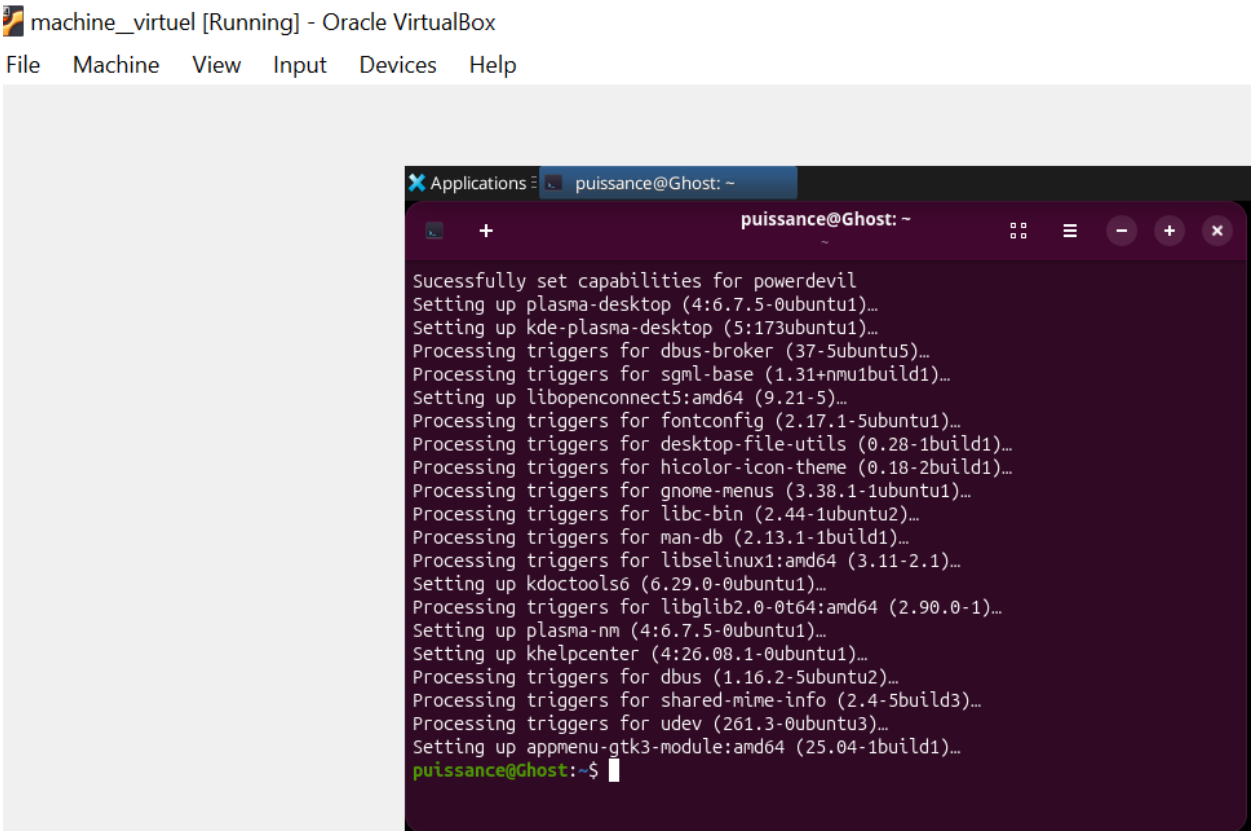
2. Disponibilité des paquets

3. Simuler séparément les installations

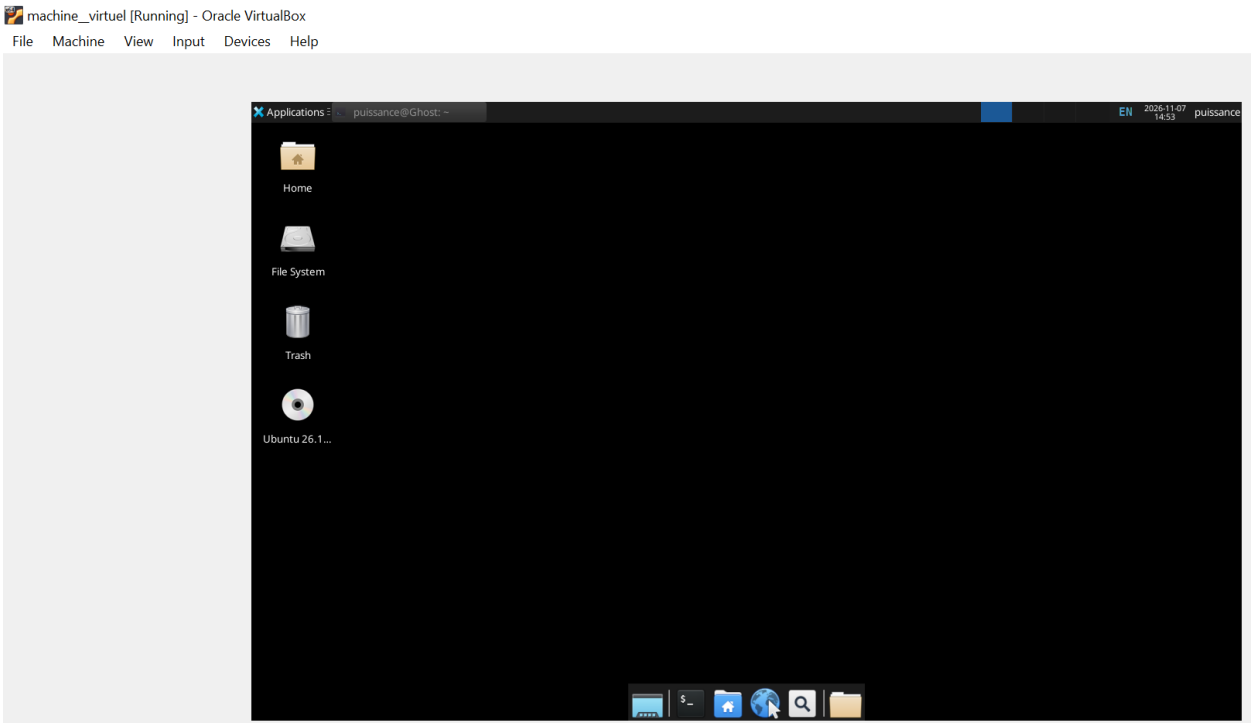


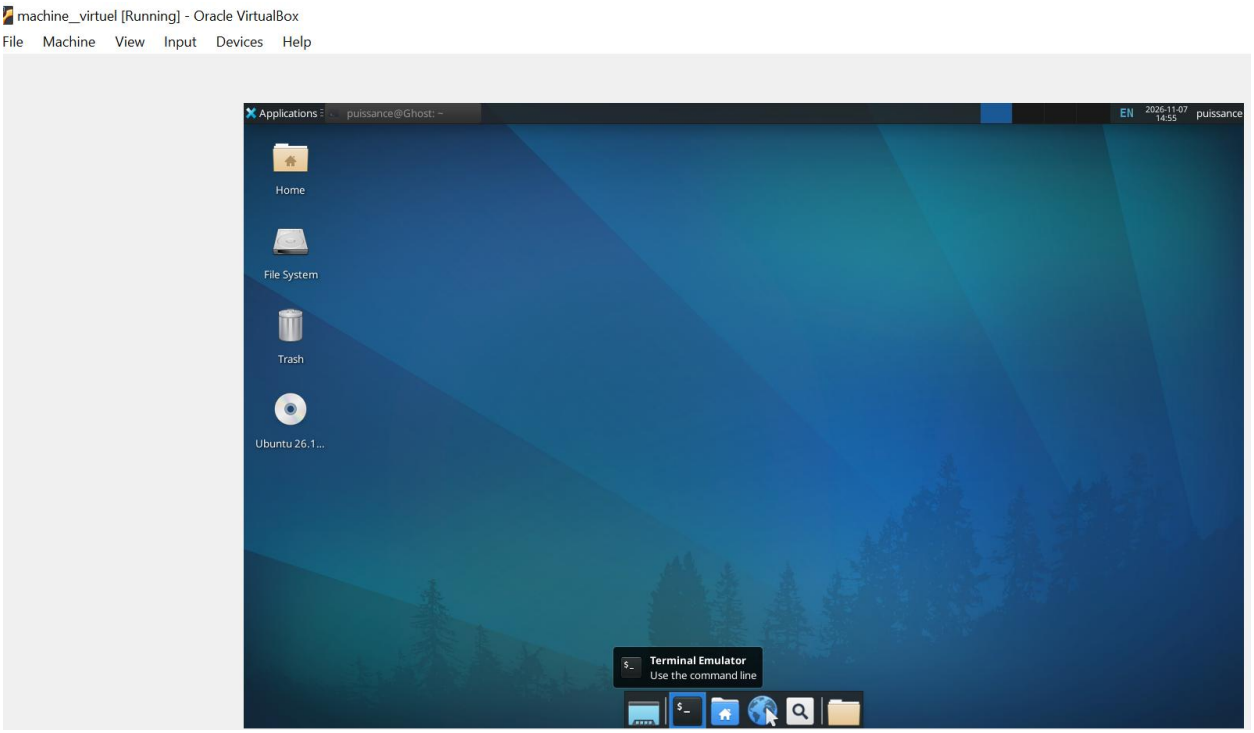
4. Installer les environnements de bureau

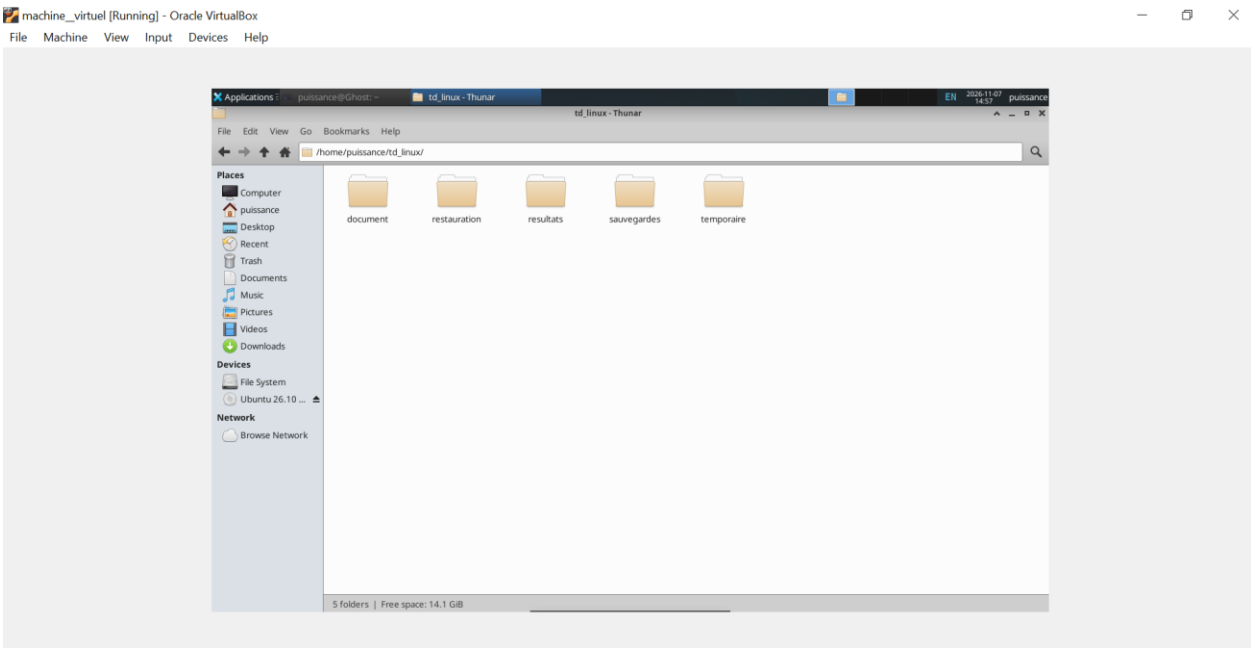


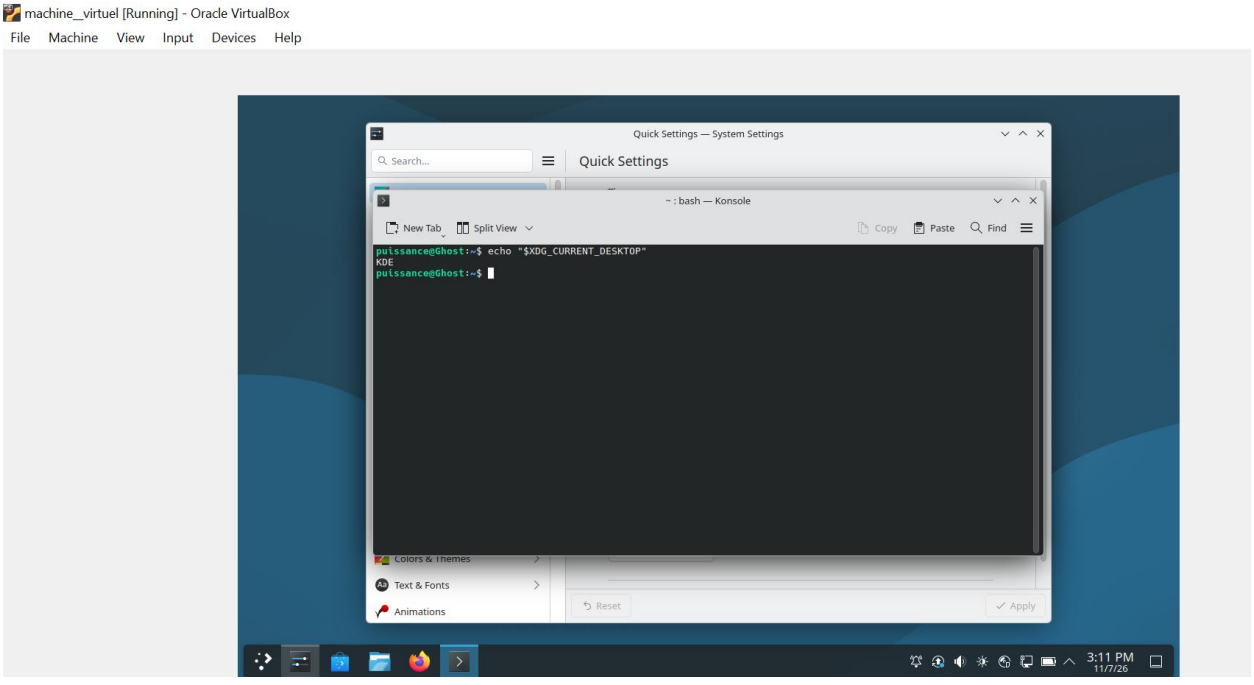


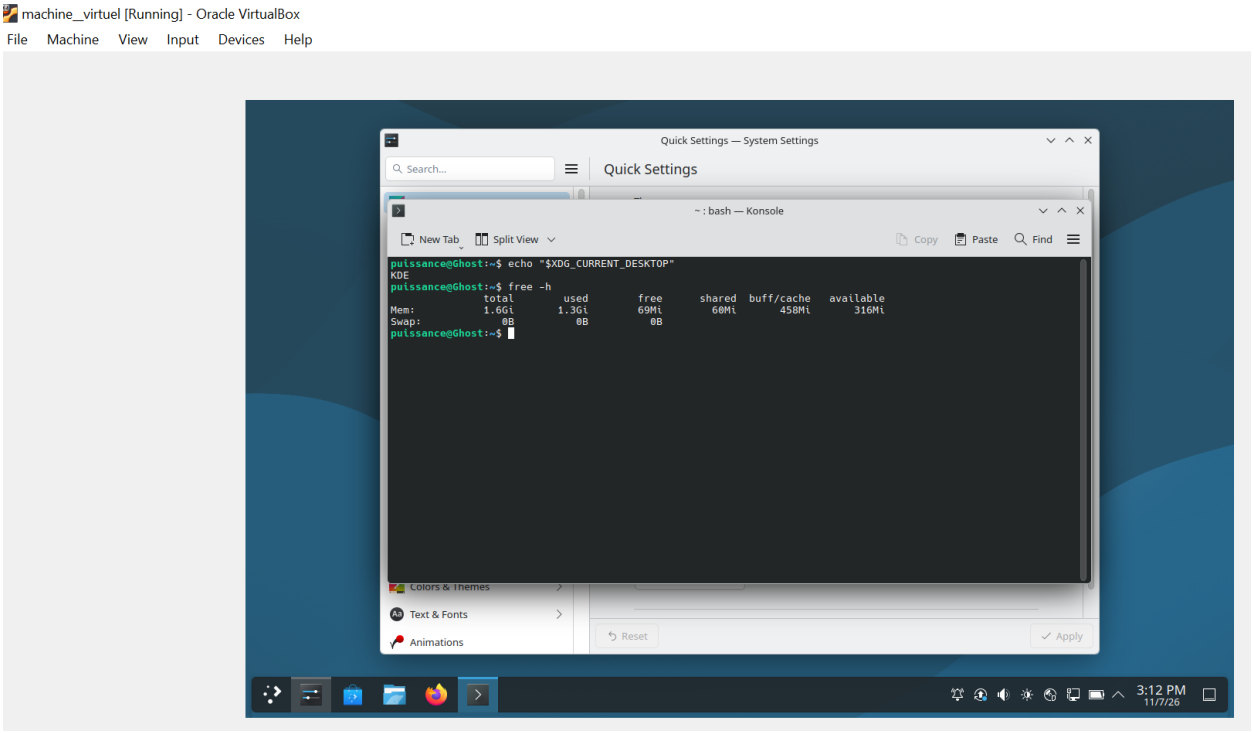
5 et 6. Tester chaque environnement de bureau

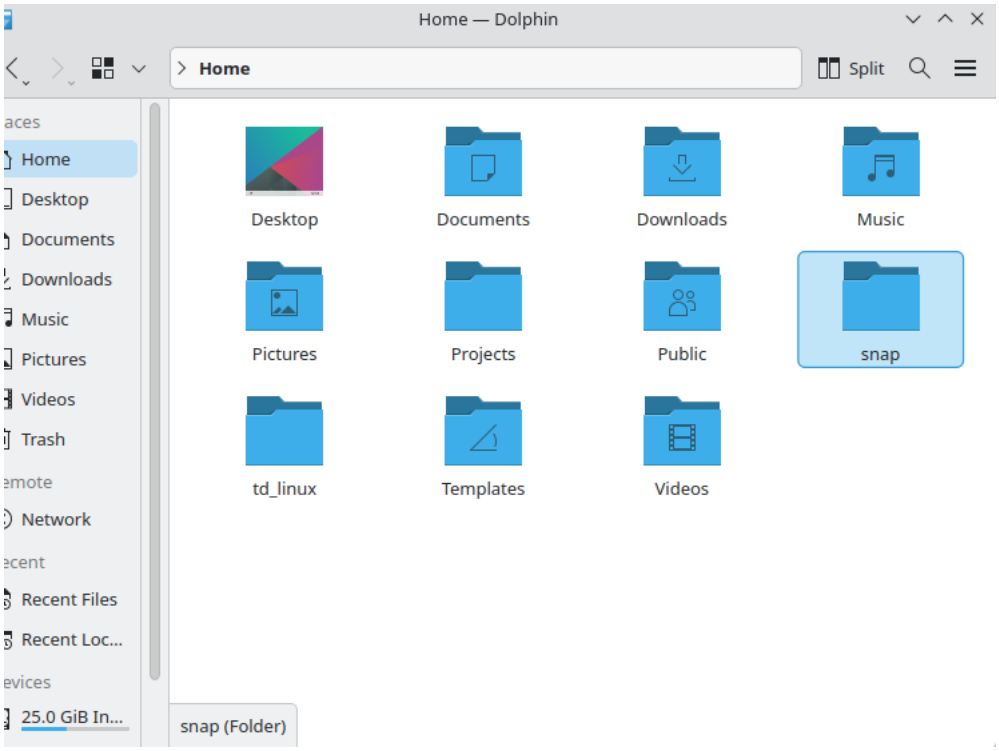





machine_virtuel [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Applications
td_linux - Thunar
File Edit View Go Bookmarks Help
/home/puissance/td_linux/
Places
Computer
puissance
Desktop
Recent
Trash
Documents
Music
Pictures
Videos
Downloads
Devices
File System
Network
Browse Network
document
restauration
resultats
sauvegardes
temporaire
5 folders | Free space: 14.1 GiB

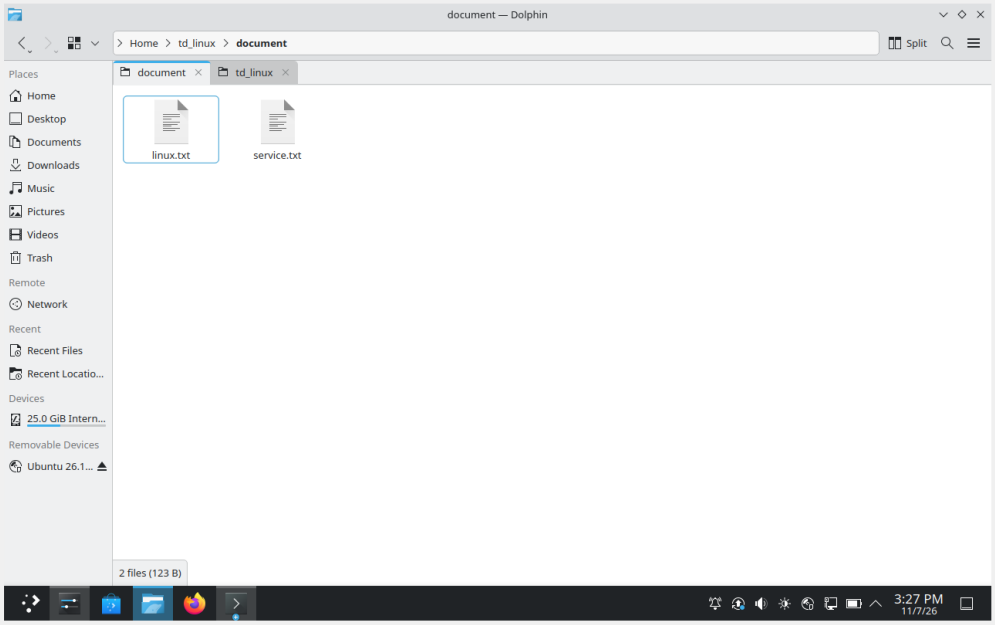

machine_virtuel [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Quick Settings — System Settings
Search...
Quick Settings
~ : bash — Konsole
New Tab
Split View
Copy
Paste
Find
puissance@Ghost:~$ echo "$XDG_CURRENT_DESKTOP"
KDE
puissance@Ghost:~$
Colors & Themes
Text & Fonts
Animations
Reset
Apply
3:11 PM
11/7/26


machine_virtuel [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Quick Settings — System Settings
Search...
Quick Settings
~ : bash — Konsole
New Tab
Split View
Copy
Paste
Find
puissance@Ghost:~$ echo "$XDG_CURRENT_DESKTOP"
KDE
puissance@Ghost:~$ free -h
total used free shared buff/cache available
Mem: 1.6Gi 1.3Gi 69Mi 60Mi 458Mi 316Mi
Swap: 0B 0B 0B
puissance@Ghost:~$
Colors & Themes
Text & Fonts
Animations
Reset
Apply
3:12 PM
11/7/26


Home — Dolphin
Home
Split
Home
Desktop
Documents
Downloads
Network
Recent Files
Recent Loc...
25.0 GiB In...
Desktop
Documents
Downloads
Music
Pictures
Projects
Public
snap
td_linux
Templates
Videos
snap (Folder)

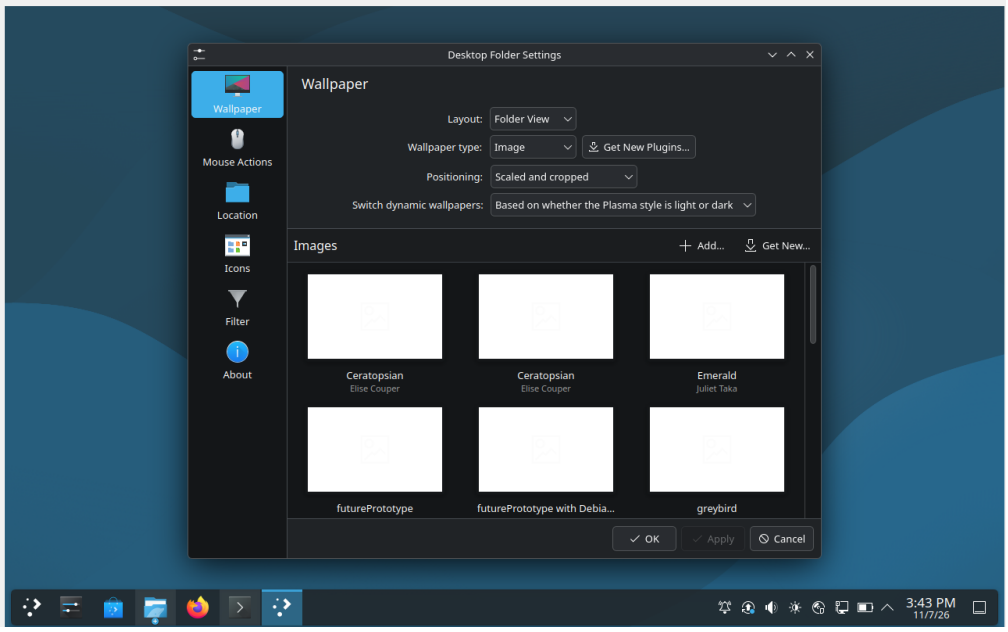
machine_virtuel [Running] - Oracle VirtualBox

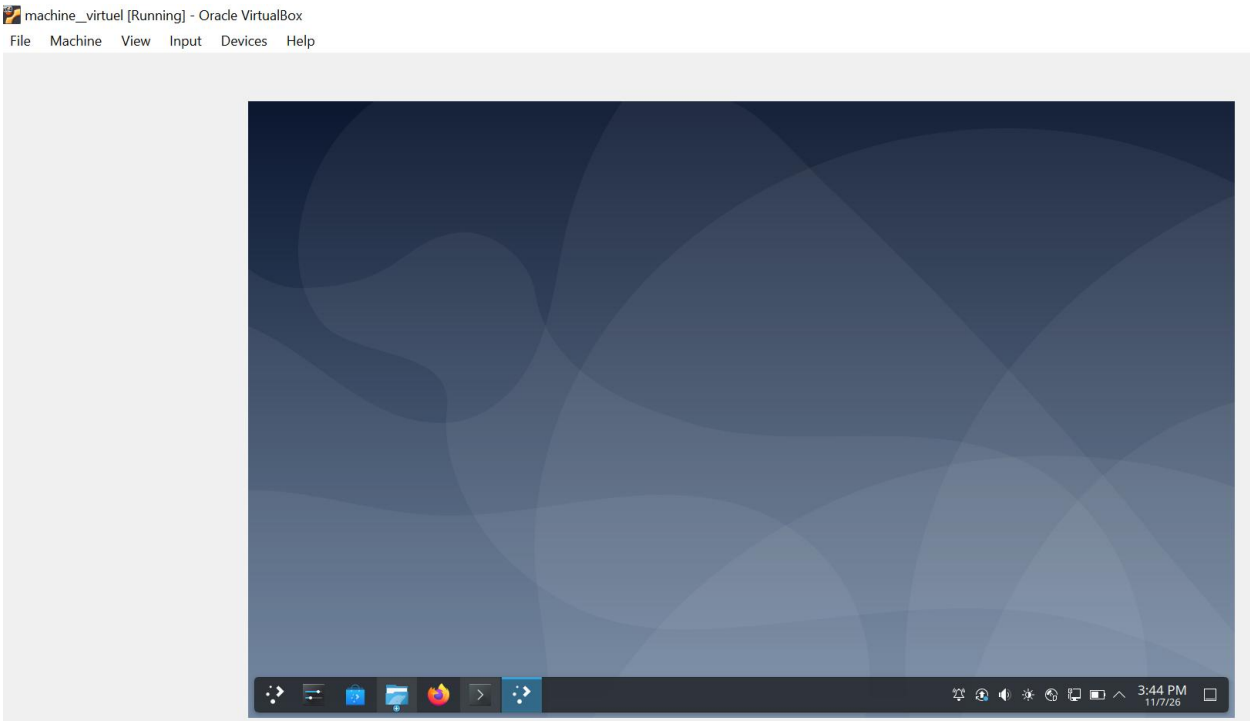
File Machine View Input Devices Help



machine_virtuel [Running] - Oracle VirtualBox

File Machine View Input Devices Help




machine_virtuel [Running] - Oracle VirtualBox
File Machine View Input Devices Help
3:44 PM
11/7/26


Paramètres
Personnel
Apparence
Bureau
Espaces de travail
Gestionnaire de fenêtres
Notifications
Peaufinage des fenêtres
Tableau de bord
Thunar Préférences
Matériel
Affichage
Clavier
Contrôle du volume
Disques et médias amovibles
Profils de couleur
Souris et pavé tactile
Système
Accessibilité
Applications par défaut
À propos de Xfce
Aide
Tous les paramètres
Fermer

7. Compléter le tableau comparatif

Critère	KDE Plasma	XFCE
Terminal utilisé	Konsole	Konsole
Gestionnaire de fichiers	Dolphin	Thunar
Accès aux paramètres	Menu des applications Configuration du système (une seule fenetre avec rechercher)	Application > parametre > Gestion de parametres (icones par categorie)
Personnalisation	Très poussée : themes,widgets , panneaux ,effet ; le clic droit sur le bureau ouvre la configuration du fond d'ecran	Plus simple, moins d'options ; parametres du bureau, apparence, panneaux
Mémoire	1,6 Gi disponibles(3,3 Gi au total ; 1,7 Gi utilises)	2,3 Gi disponibles (3,3 Gi au total ; 1,0 Gi utilisés)
Choix pour un poste donné	Poste recent avec assez de RAM (8 Go ou plus) et utilisateur voulan un bureau moderne et tres responsable	Poste ancien ou a faible RAM, Rechercher de légèreté et de rapidité

La tâche a été réalisée avec succès. Certaines commandes demandaient de l'attention pour éviter les erreurs, mais leur exécution étape par étape m'a permis de mieux les comprendre et de trouver les solutions nécessaires. Ce travail m'a donc permis d'améliorer mes connaissances pratiques sur le système Ubuntu et de gagner en expérience dans l'utilisation de Linux.

Conclusion

Ce TD m'a permis de mieux comprendre le fonctionnement de Linux et d'apprendre à utiliser plusieurs commandes dans le terminal. J'ai appris à créer, supprimer et organiser des fichiers et des dossiers, à gérer les logiciels avec APT et à découvrir différents environnements de bureau.